DA2402 Data Curation and Visualisation Project

State Health Spending and Health Outcomes

Do states that spend more per capita on health
have better maternal and child health outcomes?

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1 Introduction

1.1 Background

Mothers and young children are among the most vulnerable groups in any population. Pregnancy and childbirth carry risks for women, and children under five are especially susceptible to infection, undernutrition and anaemia, while depending entirely on others for their care. Their health also matters beyond the present: children are the future of the country, and poor health and nutrition in early childhood can affect their growth, learning and productivity for the rest of their lives. Maternal and child health is therefore one of the clearest measures of how well a health system works. Deaths of infants and young children, child undernutrition and anaemia among mothers are largely preventable through services that public health spending pays for, such as antenatal care, institutional deliveries, immunisation and nutrition programmes.

In India, health is a state subject: state governments provide most public health spending, and the amount they spend varies widely. In 2019-20, health spending per capita ranged from about Rs. 500 in some states to over Rs. 6,000 in others. Maternal and child health outcomes vary just as much: infant mortality in NFHS-5 (2019–21) ranges from 4 deaths per 1,000 live births in Kerala to 50 in Uttar Pradesh. This raises a natural question: do the states that spend more achieve better outcomes for mothers and children?

The answer is not obvious. Higher spending may not translate into better outcomes if it is spent inefficiently, does not reach rural areas, or is outweighed by other factors such as women's education, sanitation and income. A state may also spend more simply because it is small, mountainous or sparsely populated, and its people are therefore more expensive to reach.

1.2 Research Question

The main question asked is:

> *Do Indian states that spend more per capita on health have better maternal and child health outcomes?*

To answer this question, it is broken down into four parts, which correspond to the tabs of the dashboard (Section 4):

1. How much does each state spend on health per capita, and did it spend what it budgeted?
2. How healthy are mothers and children in each state, and how much care do they receive?
3. Are outcomes and care better in states that spend more?
4. Does the association survive other explanations, unusual states and the timing of the survey?

Spending is measured per capita, rather than in total, because total spending mainly reflects the size of a state. Outcomes are measured with NFHS-5 indicators of child survival, child nutrition and mothers' health, and the care received with indicators of antenatal care, institutional births, postnatal care and vaccination.

1.3 Objectives

The steps taken to answer the question were:

1. To compile state health expenditure for 2019-20 (RBI), population projections for 2020 (National Commission on Population) and NFHS-5 indicators for 29 states from publicly available sources (Section 2).
2. To clean and combine the data sets so that every state's spending, population and health indicators can be compared consistently (Section 3).
3. To calculate health spending per capita and to summarise maternal and child health in two composite indices, one for outcomes and one for care received (Section 4.2).

4. To present the relationship between spending and health in an interactive dashboard, and to measure it with rank-based correlations (Section 4).
5. To test whether the association holds after allowing for other factors, without the north-eastern states and for the states surveyed before COVID-19, and to state its limitations without interpreting it as causal (Sections 4.9 and 4.11).

2 Data Sources and Description

This project uses the following data sources to examine the relationship between state-level health expenditure and health outcomes in India.

2.1 RBI State Finances: A Study of Budgets

To obtain state-level health expenditure figures, we used the Reserve Bank of India's annual publication, *State Finances: A Study of Budgets*, available at `https://rbi.org.in/Scripts/AnnualPublications.aspx?head=State+Finances+%3A+A+Study+of+Budgets`. This is RBI's official compilation of state government budgets, reported sector-wise and across three stages of estimation for each financial year: Budget Estimates (BE), Revised Estimates (RE), and Actuals (Accounts). We downloaded the relevant files manually from RBI's publication page, since the data is provided as individual Excel files rather than through an API.

From Appendix II of this report, *Revenue Expenditure of States and Union Territories with Legislature*, we used the expenditure figure listed under **"Medical and Public Health"** (item 2 under Social Services). We kept Water Supply and Sanitation separate, since sanitation coverage is one of the outcome-side indicators we examine from NFHS-5; merging it into the spending figure would have let health spending partly explain itself.

We focus on the financial year 2019-20, as this overlaps more closely with the NFHS-5 survey period (2019–21). No single edition of the RBI report contained an audited Accounts figure for 2019-20 at the time of its own publication, so two editions were used together:

- An earlier edition reported 2018-19 (Accounts), 2019-20 (Budget Estimates), 2019-20 (Revised Estimates) and 2020-21 (Budget Estimates).
- A later edition, published the following year, reports 2019-20 (Accounts), 2020-21 (Budget Estimates), 2020-21 (Revised Estimates) and 2021-22 (Budget Estimates).

The 2019-20 Accounts figure was taken from the later edition, so that our analysis uses audited actual spending rather than an estimate. All three values for 2019-20 – accounts, budget estimates, and revised estimates – were retained in our combined data set rather than collapsing into a single figure, since the gap between planned and actual spending may itself be informative when discussing limitations of the analysis. Expenditure values are reported in Rs. Lakh.

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2.2 NFHS-5 State Indicators

The health outcome indicators were taken from the National Family Health Survey, Round 5 (NFHS-5, 2019–21), using the state-level indicator dataset available at `https://github.com/pratapvardhan/NFHS-5`, downloaded as a CSV file. NFHS-5 is India's largest demographic and health survey, and its indicators are already expressed as percentages of the appropriate target group; for example, institutional births are reported as a percentage of births, not of the total population, so no further

population-based adjustment was required for these variables.

We retained a broad set of 53 indicators from NFHS-5, each with its Urban, Rural, and Total breakdown preserved, grouped into the following categories:

- **Demographic context:** population below age 15, sex ratio, sex ratio at birth.
- **Civil registration and basic services:** birth registration, improved drinking water, improved sanitation, iodised salt use, pre-primary school attendance. Drinking water and sanitation were included as control variables, allowing us to check whether favourable health outcomes reflect general infrastructure rather than specific health expenditures.
- **Reproductive health:** indicators of adolescent pregnancy and fertility.
- **Mortality:** neonatal, infant and under-five mortality rates – the standard outcome measures for state-level health system performance.
- **Maternal and delivery care:** antenatal visits, institutional births, skilled birth attendance, and postnatal care, chosen as process outcomes closest to direct public health spending.
- **Immunisation:** full vaccination coverage and individual vaccine doses, which depend on service delivery and, therefore, expected to respond directly to public spending.
- **Childhood diseases and feeding practices:** prevalence of diarrhoea paired with treatment-seeking behaviour and breastfeeding indicators.
- **Child nutrition:** prevalence of stunting, wasting, and underweight.
- **Financial protection:** out-of-pocket expenditure per delivery and health insurance coverage, included as a direct test of whether public spending reduces household health costs.
- **Anaemia:** prevalence among children, pregnant women, and all women.
- **Women's health and education:** literacy, years of schooling, and BMI, included as control variables for possible confounding.

This set was collected broadly at the data collection stage so that a representative subset could be chosen for visualisation, rather than limiting the scope of the collection itself.

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2.3 Population Projections for India and States, 2011–2036

Population estimates were obtained from the *Population Projections for India and States 2011–2036*, published by the National Commission on Population, Ministry of Health and Family Welfare, available at `https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf`. This report provides projected population figures for every Indian state, broken down by sex and by reference date within each year.

We used Table 8 of this report, *Projected Total Population by Sex as on 1st March*, taking only the **Persons** column (not the Male or Female sub-columns), for the year 2020, to pair with our FY 2019-20 expenditure figures. We used total population rather than a gender- or age-restricted figure because RBI's health expenditure data is not disaggregated by sex or age: since the numerator (total health spending) covers the whole population, the denominator was chosen to match it. Population figures in the source are reported in thousands and were multiplied by 1,000 during collection to obtain actual headcounts before computing per capita expenditure.

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used here are unmodified and attributed to NHM accordingly.

2.4 Summary of Data Sources

Source	Format	Access	Fields used	Period
RBI State Finances	XLSX	Manual download	Medical and Public Health expenditure - Accounts, BE, RE (Rs. Lakh)	FY 2019-20
NFHS-5	CSV	GitHub download	53 indicators across mortality, maternal care, immunisation, nutrition, sanitation, financial protection, anaemia, and women's health (Urban/Rural/Total each)	2019–21
NHM Population Projections	PDF	Manual download	Table 8, Persons column, 1st March row	2020

Table 1: Summary of data sources, access methods, and fields used

3 Data Cleaning

3.1 Overview

The combined file (`budget_nfhs_combined.csv`), built from the three sources in Section 2, stores every NFHS-5 indicator across Urban, Rural, and Total columns. It uses a two-row header, with the indicator name merged across three columns in the top row and the Urban, Rural, and Total labels sitting in the row below it. The raw file is loaded as **31 rows by 182 columns** (two header rows plus 29 states). Reading it with pandas' default header handling would silently misread this structure, so the header was first reshaped. The remaining steps, type correction, duplicate checking, outlier detection and missing value handling, were then carried out and justified individually, as described below.

3.2 Data Type Correction and Feature Engineering

Since the file was loaded with no header to preserve the Urban, Rural and Total's structure, every column was read in as text. So numeric columns was coerced to numeric dtype. Additionally, the four budget and population columns additionally use the Indian numbering system with commas, for example `6,76,767`, which was stripped before casting to float. After this step the table held 151 float64 columns, 12 int64 columns, and one object column (`State`).

Since `Accounts`, `Budget Estimates` and `Revised Estimates` are reported in Rs. Lakh as absolute state totals, a larger state will always out spend a smaller one in absolute terms regardless of how much it spends per capita. Therefore, three per capita expenditure columns were engineered as

$$\text{Per Capita Expenditure} = \frac{\text{Value in Lakh} \times 10^5}{\text{Population (Total)}}.$$

This brought the working table to 167 columns.

3.3 Duplicate Check

A row level and state name check confirmed there were **no fully duplicated rows** and no duplicate state names: 29 rows for 29 unique states.

3.4 Outlier Detection

Five complementary methods were used to test to find the outliers. **IQR**, the **modified Z-score** (median and median absolute deviation based, threshold 3.5) and **Grubbs' test** (two sided, iterative, $\alpha = 0.05$) each examine one column at a time. **DBSCAN** and **Local Outlier Factor** examine all columns jointly. However, since 166 standardised numeric columns against only 29 states makes raw distance unreliable, both were run on the principal components explaining 90% of the variance (14 components), with DBSCAN's ϵ chosen from a k distance elbow plot.

Method	Setting	Result
IQR	$[Q1 - 1.5\,IQR,\ Q3 + 1.5\,IQR]$	up to 5 flagged values per column (worst: child overweight indicators)
Modified Z-score	median/MAD, threshold 3.5	up to 5 flagged values per column, including all three per capita expenditure columns
Grubbs' test	two sided, $\alpha = 0.05$	up to 4 flagged values per column (per capita BE, BCG coverage)
DBSCAN	PCA to 14 components (90% variance), $\epsilon = 12.0$, min_samples= 3	6 states flagged as noise
Local Outlier Factor	same PCA space, $k = 3$	0 states flagged

Table 2: Outlier detection methods and their results.

The six states flagged by DBSCAN were Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Sikkim, also appeared in at least one of the column level (IQR, Z-score or Grubbs) flags, so the multivariate and univariate findings agree on which states stand out.

However, no point was removed or capped. Because, these states are genuinely small hill and north eastern states with a different population scale and survey profile from the larger plains states, so every flagged value is a plausible, in range figure.

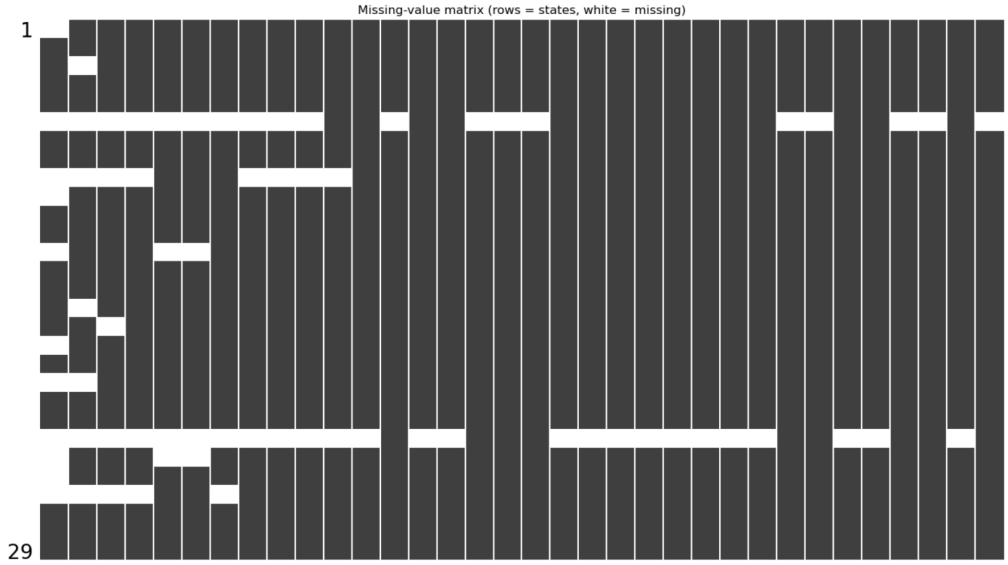
3.5 Missing Value Analysis

Indicator family	Missing (states)	% missing
Home birth check up within 24 hours (Urban/Rural/Total variants)	4 to 11	13.8 to 37.9
Non-breastfeeding children 6–23 months, adequate diet	3 to 7	10.3 to 24.1
Children 6–8 months on solid/semi-solid food and breastmilk	5	17.2
Children under 6 months exclusively breastfed	4	13.8
Neonatal, infant and under-five mortality rate (Urban)	3 each	10.3 each
Pregnant women anaemic	2	6.9
Remaining 20+ columns	1 each	3.4 each

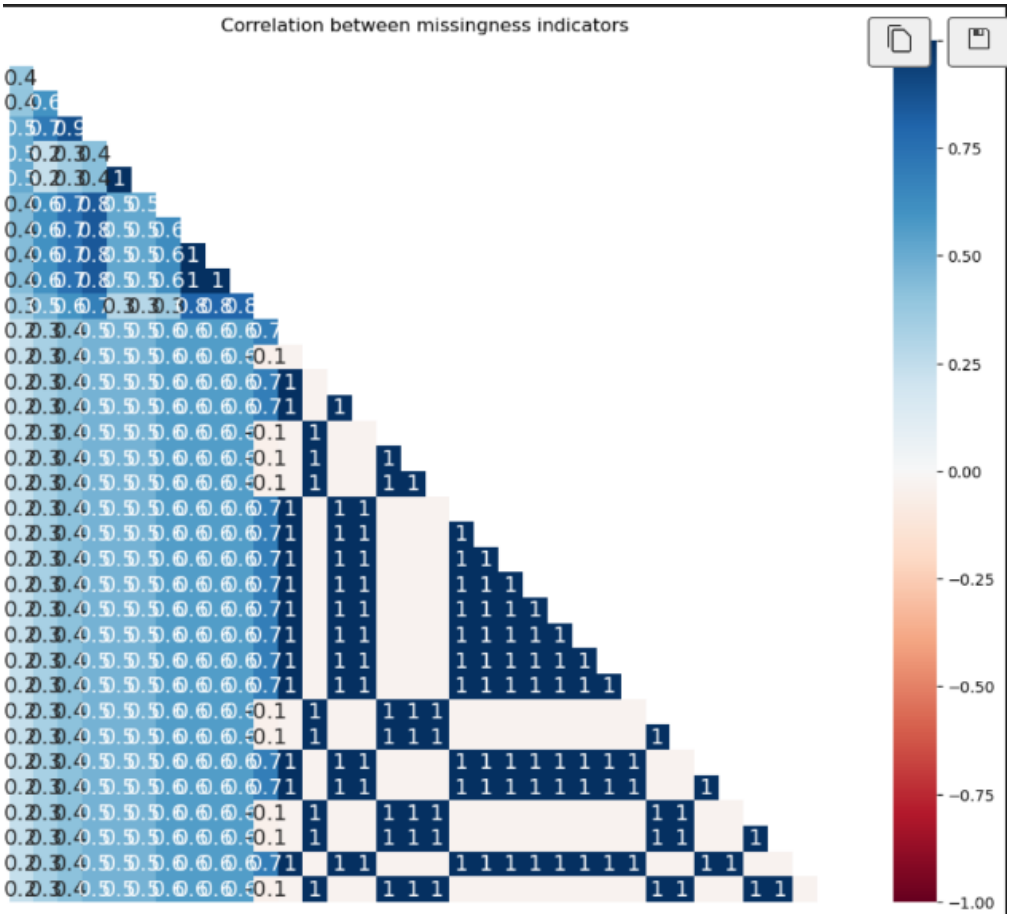
Table 3: Columns with missing values, grouped by indicator family.

34 of the 167 columns had at least one missing value out of 29 states. Table 3 summarises the most affected indicator families.

A missingno matrix and co-missingness heatmap (Figure 1) were used to check whether gaps occur independently or together. Most of the Urban columns are missing together for the same small states, consistent with those states having little or no urban population to report on, rather than random data loss.



(a) Missing value matrix



(b) Co-missingness heatmap

Figure 1: Missingness pattern across states and co-missingness between indicators.

Little's global MCAR test was also run, returning $p \approx 0.0000$, rejecting the null hypothesis that

missingness overall is unrelated to the observed values. With 166 numeric columns against only 29 rows, there are far more parameters than data points, so this result is read with caution rather than as firm evidence.

3.6 Imputation

Mean and median imputation were deliberately avoided. Bcause in our case, each row is one state's health profile for one reference year, not a repeated sample, so replacing a missing value with an across state typical figure would erase exactly the state level variation this project studies.

Instead, a correlation guided K nearest neighbours approach was used for every missing column. For each column, the six other numeric columns it correlates with most strongly (absolute Pearson correlation, computed pairwise on the available data) were identified, and `KNNImputer` ($k = 5$) was run on the standardised block made up of that column plus its top correlated features only, rather than the full 166 column space. Columns were processed in ascending order of missing count, so columns imputed later could benefit from features already filled in by earlier steps.

All 34 affected columns were fully imputed in this way, leaving zero missing values in the final 29 by 167 table. Figure 2 compares the original and imputed distributions for a sample of the affected columns.

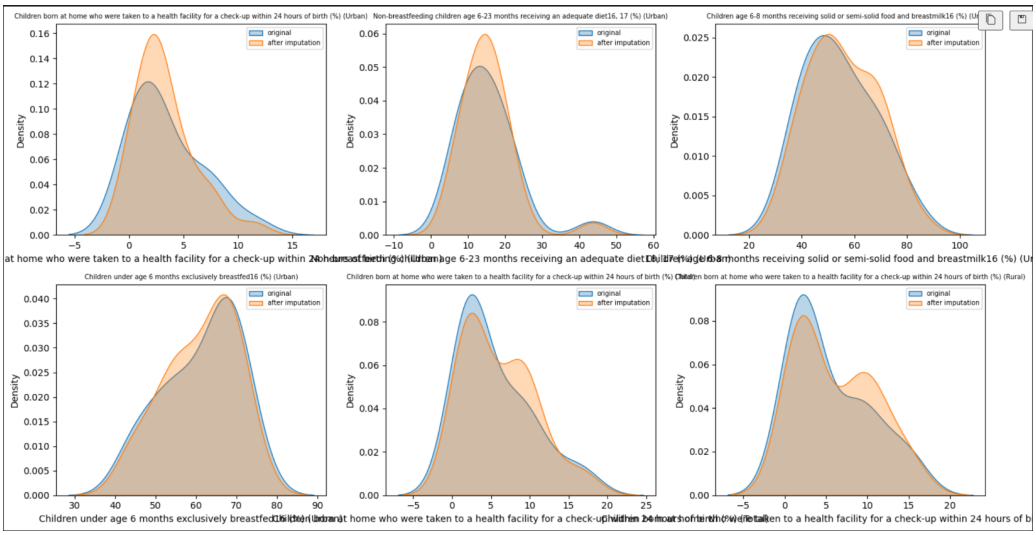


Figure 2: Original versus imputed distributions for a sample of affected columns.

The cleaned dataset was saved as `budget_nfhs_combined_clean.csv`.

4 Data Visualisation

4.1 Approach

The cleaned data was visualised as an interactive dashboard built with Plotly Dash. This dashboard is organised into six tabs. Each section addresses a part of the question, moving from an overview of spending to health outcomes and finally the relationship between them.

The six tabs in the dashboard are:

1. **Spending Landscape:** how much each state spends on health per capita, how unevenly and whether it spent what it budgeted.
2. **Health Outcomes:** the health outcomes of mothers and children in each state.
3. **Spending vs Outcomes:** whether states that spend more have better outcomes.
4. **Is it really the spending?:** whether the link survives other explanations, outlier states and the timing of the survey.

5. **Data Quality:** the data quality issues identified during cleaning (Section 3) and how they were resolved.
6. **Findings and Methods:** the key findings, how the analysis was performed and its limitations.

Each tab begins with four headline figures, followed by a series of charts. Each chart is produced by a single Python function in `figures.py`, which returns both the figure and its caption. All numerical values in the captions are calculated directly from the data, keeping them consistent with the charts. The same functions are used in both the dashboard (`app.py`) and the visualisation notebook (`data_visualisation.ipynb`).

Unless stated otherwise, all charts use the **Total** (state-wide) NFHS-5 figures for the 29 states ($n = 29$). Urban and Rural figures are used only where explicitly indicated.

4.2 Derived Measures

The definitions of the quantities used in the charts and the rationale for their selection are provided below.

Spending per capita

The Reserve Bank of India reports state health expenditure as an absolute amount in Rs. Lakh (1 Lakh = 10^5). Since larger states will naturally have higher total expenditure, these figures mainly reflect differences in state size. To allow meaningful comparisons, expenditure is expressed per capita, calculated for state i as

$$\text{pc}_i \;=\; \frac{\text{Accounts}_i \times 10^5}{\text{Population}_i} \quad \text{(rupees per person per year)}.$$

Here, Accounts_i represents the audited actual expenditure on Medical and Public Health in 2019-20, while Population_i is the projected population as of 1 March 2020 (Section 2.3). The same formula is applied to the Budget Estimate and the Revised Estimate, giving the corresponding budgeted and revised per capita spending. Actual expenditure (Accounts) is used throughout the analysis because it reflects the amount actually spent rather than the amount planned.

For the treemap, the box size represents each state's total health spending, while its colour indicates the per capita spending. The total expenditure is converted from Rs. lakh to Rs. crore (1 crore = 100 lakh) for readability.

Budget utilisation

Budget utilisation measures how much of the allocated budget was spent by a state in reality. It is calculated as

$$\text{Utilisation}_i \;=\; \frac{\text{Accounts}_i}{\text{Budget Estimate}_i} \times 100\%.$$

A value below 100% indicates that a state has spent less than it has budgeted, while a value above 100% means that the actual expenditure has exceeded the original budget estimate.

Indicator scores (0-100)

NFHS-5 indicators are reported in different units, such as deaths per 1000 live births and percentages of children or women. They also differ in direction: for instance, lower mortality is better, while higher vaccination coverage is preferred. In order to make them comparable and allow them to be combined, each indicator was rescaled across the 29 states using min-max scaling:

$$s_i \;=\; \begin{cases} \dfrac{x_i - \min(x)}{\max(x) - \min(x)} \times 100, & \text{if a higher value is better,} \\[2ex] 100 - \dfrac{x_i - \min(x)}{\max(x) - \min(x)} \times 100, & \text{if a lower value is better.} \end{cases}$$

As a result, every indicator now has a common scale, where 0 represents the worst-performing state and 100 the best, regardless of the original unit or direction.

For example, infant mortality ranges from 4.4 deaths per 1000 live births in Kerala to 50.4 in Uttar Pradesh. Since lower mortality is better, Kerala receives a score of 100 and Uttar Pradesh a score of 0. Bihar, with an infant mortality rate of 46.8, scores

$$100 - \frac{46.8 - 4.4}{50.4 - 4.4} \times 100 = 7.8.$$

These scores are *relative*: a score of 100 means that a state performs best among the 29 states, not that it has reached an ideal level for that particular indicator.

Composite indices

To summarise the indicators, two composite indices were constructed by taking the simple average of their respective indicator scores:

$$\text{Outcome Index}_i = \frac{1}{7}\sum_{k=1}^{7} s_{ik}, \qquad \text{Care Index}_i = \frac{1}{6}\sum_{k=1}^{6} s_{ik}.$$

- **Outcome Index** (7 indicators): infant mortality, under-five mortality, stunting, wasting, underweight, child anaemia and anaemia in pregnancy. These indicators capture different aspects of maternal and child health.
- **Care Index** (6 indicators): antenatal check-up in the first trimester, four or more antenatal visits, iron-folic acid supplementation for 180 or more days during pregnancy, institutional births, postnatal care within two days and complete vaccination. These indicators cover the key stages of maternal and child healthcare, from pregnancy to early childhood, and reflect services closely linked to public health provision.

Both indices range from 0 to 100. Equal weights were assigned to all indicators as there is no established basis for according greater importance to one indicator over another.

The heat maps and correlation chart cover a wider set of individual indicators (10 outcome indicators and 9 care indicators), while the composite indices utilise only the core subsets listed above. Neonatal mortality is excluded from the Outcome Index since neonatal deaths are already included in both infant and under-five mortality. Including it separately would therefore give additional weight to deaths already captured by these indicators. Similarly, anaemia among all women substantially overlaps with anaemia in pregnancy.

Feeding practices, such as exclusive breastfeeding and adequate diet, along with Vitamin A supplementation are excluded from the Care Index because they are influenced by household practices and nutritional programmes, rather than being part of the core maternal and child healthcare pathway.

4.3 Statistical Methods

Logarithmic Transformation and Rank-Based Correlation

Per capita health spending is strongly right-skewed (skewness = 1.79). Most states spend between Rs. 500 and Rs. 1,500 per capita, although a few smaller states spend over Rs. 4,000 per capita. As a result, the mean (Rs. 1,940) is considerably higher than the median (Rs. 1,304). Thus, two choices were made:

- Spending is plotted on a **logarithmic axis**, where equal distances represent equal spending ratios. Without a log scale, most states would have been compressed towards the left of the charts.
- Relationships are examined using **the Spearman rank correlation** rather than Pearson's correlation. Spearman's correlation is based on ranks and is less sensitive to extreme values, making it more suitable for the skewed spending data.

Spearman's rank correlation

For each variable, the 29 states are assigned ranks in ascending order (1 = lowest), with tied values receiving the average of their corresponding ranks. Spearman's correlation is then calculated using these ranks:

$$\rho \;=\; \frac{\sum_i (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_i (R_i - \bar{R})^2 \, \sum_i (S_i - \bar{S})^2}},$$

where R_i and S_i denote the ranks of state i for the two variables. The coefficient ρ ranges from -1 to $+1$. A value of $+1$ indicates identical ranking in the two variables, while -1 indicates exactly opposite ranking. A value of 0, on the other hand, indicates that there is no monotonic association.

Since Spearman's correlation depends only on the relative ordering of the observations rather than on their actual values, it is far less sensitive to skewness and extreme observations than Pearson's correlation.

For the captions, the correlation strength is classified as **strong** for $|\rho| \geq 0.7$, **moderate** for $0.4 \leq |\rho| < 0.7$, and **weak** for $|\rho| < 0.4$.

Direction (signed correlations)

Some indicators are better when higher (e.g., vaccination), while others are better when lower (e.g., mortality). To make their directions comparable, raw correlations are multiplied by $+1$ for the former and -1 for the latter. Thus, a positive ρ always indicates an association between higher spending and better outcomes. For example, the raw correlation with infant mortality is $\rho = -0.57$, which becomes $+0.57$ on the signed charts.

Significance and confidence intervals

The p-value tests whether ρ differs significantly from zero, using the test statistic

$$t \;=\; \rho \sqrt{\frac{n-2}{1-\rho^2}}$$

with $n - 2$ degrees of freedom (two-sided). Results are considered **statistically significant** when $p < 0.05$.

A 95% confidence interval for ρ is calculated using a standard approximation for Spearman's correlation:

$$z = \tanh^{-1}(\rho), \qquad \mathrm{SE} = \sqrt{\frac{1 + \rho^2/2}{n-3}}, \qquad \mathrm{CI} = \big[\tanh(z - 1.96\,\mathrm{SE}),\ \tanh(z + 1.96\,\mathrm{SE})\big].$$

Confidence intervals can be wide with only 29 states. They are therefore shown alongside correlation estimates in the charts.

Partial correlation (controlling for another factor)

To examine whether the relationship between spending and outcomes remains after accounting for a third factor z (such as women's schooling), partial Spearman correlation is used. All three variables are first converted to ranks. The linear relationship of each spending and outcome rank with the rank of z is then removed, and the correlation between the remaining residuals is calculated:

$$e_x = R_x - (a_x + b_x R_z), \qquad e_y = R_y - (a_y + b_y R_z), \qquad \rho_{xy\cdot z} = \mathrm{corr}(e_x, e_y).$$

Thus, this answers the question of whether higher spending is still associated with better outcomes in states with similar values of z. The p-value is calculated using $n - 3$ degrees of freedom.

Trend lines and prediction gap

Trend lines on the scatter plots are least-squares regression lines of the outcome on the logarithm of spending:

$$\hat{y}_i = a + b \log_{10}(\text{pc}_i),$$

consistent with the logarithmic spending axis. The vertical distance between a state's observed outcome and its predicted outcome is calculated as

$$\text{gap}_i = \text{Outcome Index}_i - \hat{y}_i.$$

A positive gap indicates that a state performs better than predicted for its spending and vice versa. This is, however, a descriptive measure and not one of efficiency as the gap may also reflect factors other than how the money is used.

Groups of states

Some charts compare groups of states ranked by per capita spending: **thirds** (the top and bottom 9 states) in the heatmap captions and **quarters** (approximately 7 states each) in the grouped bar chart. These comparisons use the averages of the 0-100 indicator scores.

Robustness checks

Two subsets are used to examine whether the results are sensitive to the inclusion of particular states or survey periods:

- **Excluding the six north-eastern states** (Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Sikkim; $n = 23$), which were flagged as outliers in Section 3.
- **States surveyed before COVID-19 only**: the 18 states covered in NFHS-5 Phase 1, surveyed between June 2019 and January 2020 ($n = 18$).

4.4 Design Principles

All charts follow a consistent set of design principles:

- **Labelling:** Every chart has a title, axis labels with units, and a legend wherever colour carries meaning. Each chart also has a caption of two to three sentences highlighting its main finding with numerical values.
- **Greyscale readability:** Regions are distinguished using both colour and marker shape (for example, circles for the South and stars for the Northeast), so that charts remain interpretable in greyscale and for colour-blind readers. Continuous values use sequential colour scales with steadily changing lightness.
- **Highlight colour:** Cyan is reserved for highlighting the state selected by the reader. Statistical significance is indicated separately using filled versus hollow markers or solid versus dotted lines, so that the two are not confused.
- **Gestalt principles:** Related charts are placed next to each other within a tab (proximity). Charts intended for comparison share the same state order and scale (alignment and similarity). Each tab is also organised around a single question.
- **Dark theme:** A dark background with bright, colourful marks is used to provide contrast on screen.

4.5 Interactivity

A dropdown at the top of the dashboard allows the reader to highlight a state across all charts at once. States can also be selected by clicking them on the map or on most charts. Four charts have

their own controls: the main scatter plot (choice of indicator; Total, Urban or Rural figures; and an option to exclude the six north-eastern states), the rural–urban comparison (choice of indicator), the confounder check (choice of factor), and the before-and-after imputation plot (Total, Urban or Rural figures). All charts support hover tooltips and zooming.

The dashboard can be run locally using `python app.py`. A static version, which supports hover tooltips and zooming but not the state selector or chart controls, is published at `https://iitm-da25b042.github.io/DCV-Project-2026/`.

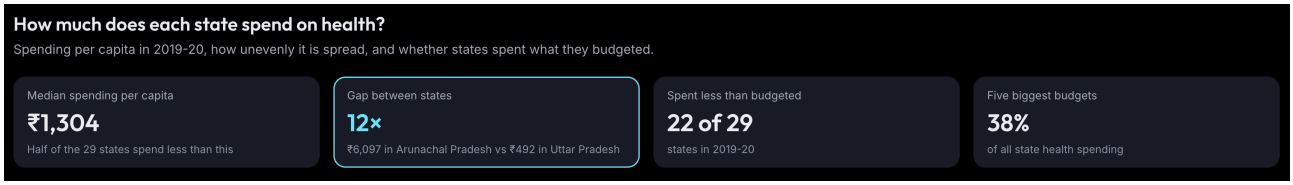


Figure 3: The dashboard showing the tab bar, headline figures and first row of charts in Tab 1.



(a) Kerala selected and highlighted in cyan.



(b) Indicator changed and north-eastern states excluded.

Figure 4: Examples of the dashboard's interactive features.

4.6 Tab 1: Spending Landscape

Before relating spending and health outcomes, Tab 1 examines health spending on its own: how much each state spends per capita, how this spending varies across states and whether states spent what they budgeted. The four headline figures at the top of this tab are:

Headline	Value	Meaning
Median spending per capita	Rs. 1,304	Half of the 29 states spend less than this
Gap between states	12×	Rs. 6,097 in Arunachal Pradesh vs Rs. 492 in Uttar Pradesh
Spent less than budgeted	22 of 29	States whose actual spending in 2019-20 was below the Budget Estimate
Five biggest budgets	38%	Share of all state health spending accounted for by the five largest budgets

Table 4: Headline figures for Tab 1.

Health spending per capita by state (choropleth map, univariate)

Chart Description: Each state is coloured according to its health spending per capita in 2019-20.

Interpretation: The colour scale runs from dark purple (lowest expenditure) to bright yellow (highest expenditure), using a logarithmic scale to distinguish differences among the many low-spending states. The colour bar below the map gives the corresponding value in rupees. States and Union Territories not included in the study (such as Delhi and Ladakh) are shown in grey.

Rationale: A map makes it easier to identify geographical patterns in spending than a table.

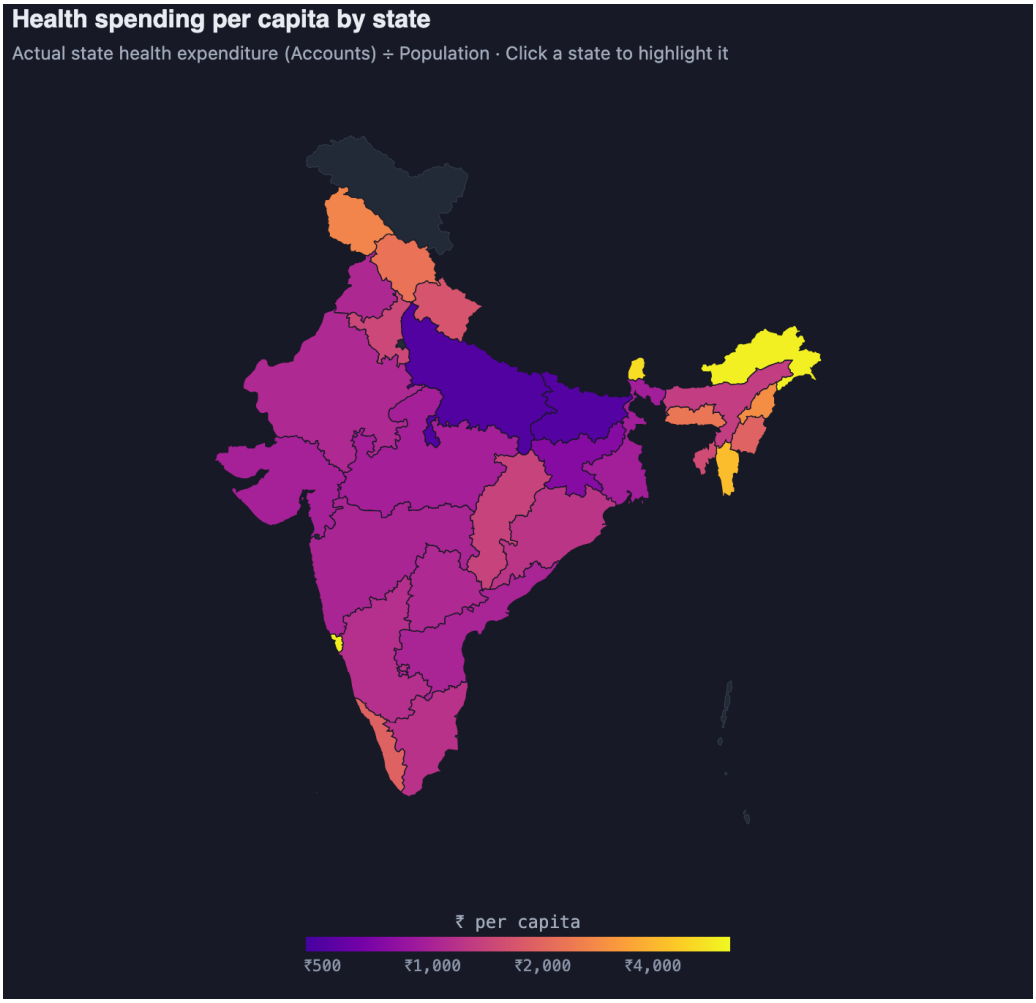


Figure 5: Per capita expenditure ranges from Rs. 492 in Uttar Pradesh to Rs. 6,097 in Arunachal Pradesh, a 12-fold gap. Many of the highest-spending states are small, hilly or north-eastern states, while the large plains states of the north and east tend to have lower per capita spending. The median is Rs. 1,304.

Health spending per capita, ranked by state (lollipop chart, univariate)

Chart Description: The same spending values as the map are sorted in descending order so that each state's position can be compared directly.

Interpretation: Each state is represented by a horizontal line (the "stick") ending in a marker at its per capita spending value. The marker's colour and shape indicate the state's region, while the dashed vertical line marks the median (Rs. 1,304).

Rationale: A lollipop chart presents the same information as a bar chart with less visual clutter, keeping all 29 states readable. Sorting makes the ranking and the difference from the median easy to see.

A large budget does not imply high per capita expenditure (slope chart, bivariate)

Chart Description: How each state's rank changes when expenditure is measured per capita rather than in total.

Interpretation: The axis on the left ranks states by total health expenditure and the right axis by spending per capita (1 = highest on both). Each line joins a state's two ranks: a line rising to the right indicates a better per capita rank than total-spending rank, while a falling line points to a worse per capita rank.

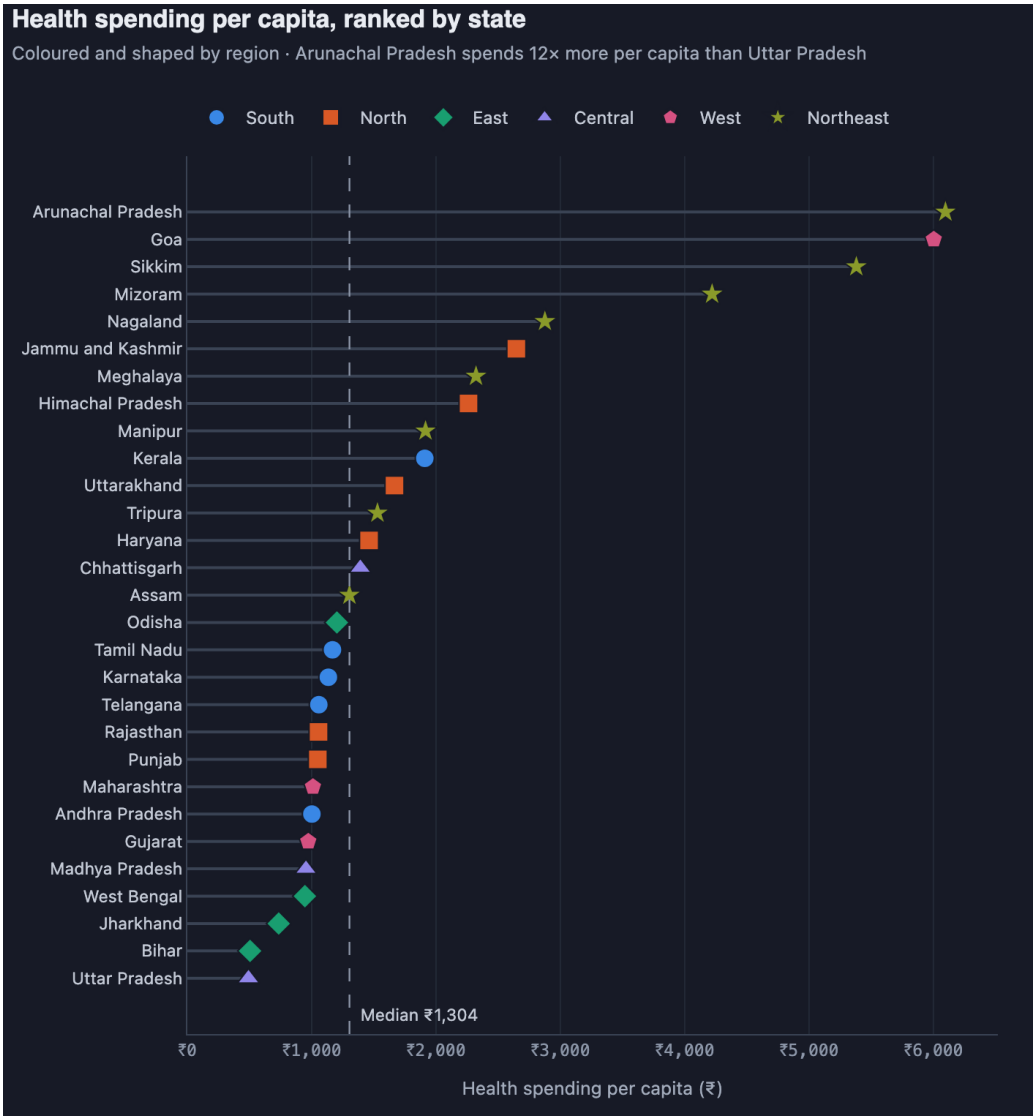


Figure 6: Half the states spend less than Rs. 1,304 per capita. All of the 8 highest spenders have populations below 15 million, and 5 of those 8 are in the Northeast. These states may face higher costs of reaching dispersed populations.

Rationale: It illustrates why states are compared on a per capita basis rather than by total spending.

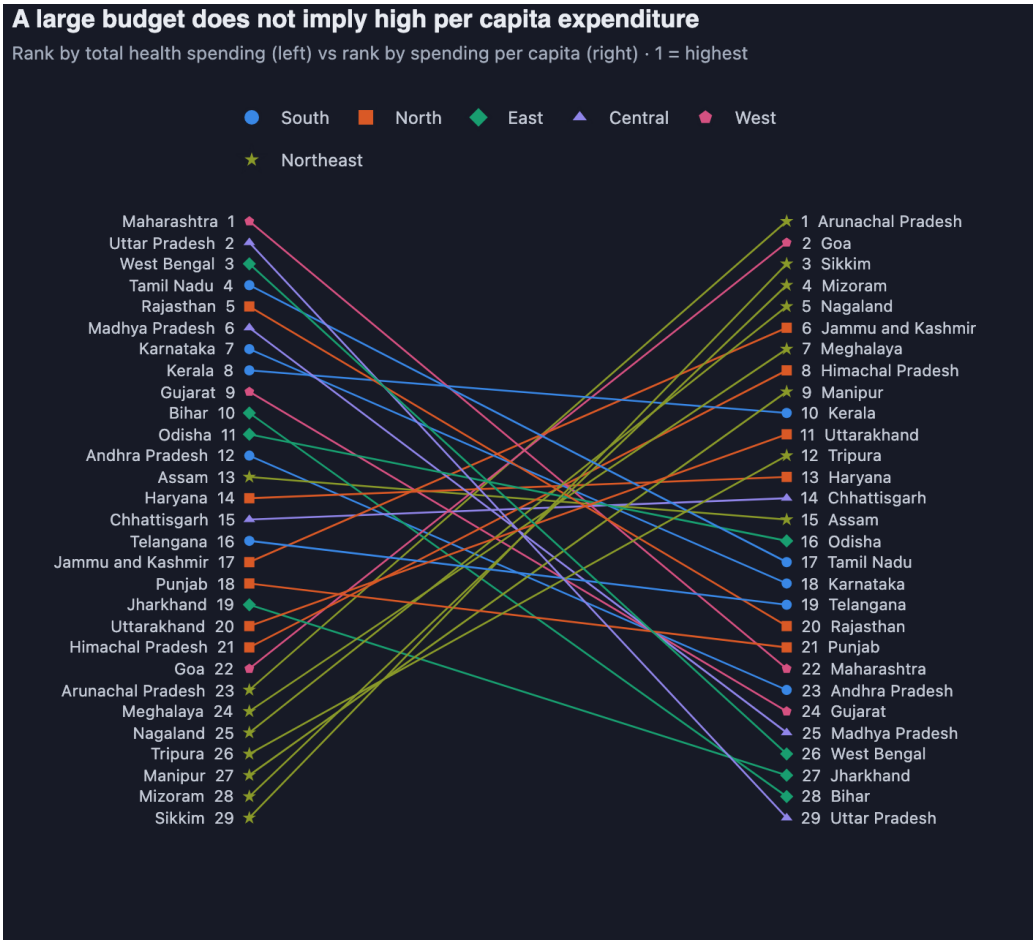


Figure 7: Uttar Pradesh has the 2nd-largest health budget in total but ranks 29th per capita, while Sikkim moves from 29th to 3rd. Total expenditure therefore reflects state size as well as spending levels, which is why spending comparisons are reported in per capita terms.

Where is the health money and how thinly is it spread? (treemap, part-to-whole)

Chart Description: Total health spending broken down by region and then by state, along with per capita spending.

Interpretation: The area of each box is proportional to the state's total health spending (in Rs. crore). The colour shows per capita spending, using the same scale as the map. A large dark box represents a large total budget but relatively low spending per capita, while a small bright box represents a smaller total budget but higher spending per capita.

Rationale: A treemap displays two dimensions at once: each state's share of total expenditure (area) and its per capita spending (colour).

The all-state figure of Rs. 988 is total spending divided by total population, so it is weighted towards more populous states. It is therefore lower than the median state value of Rs. 1,304.

Most states cluster at the low end (histogram, univariate)

Chart Description: The distribution of per capita spending across the 29 states.

Interpretation: Each bar counts the states in a Rs. 500 band of per capita spending. Brighter bars represent the bands containing more states, as indicated by the colour bar. The small ticks below the axis mark individual states, the dashed line the median and the dotted line the mean.

Rationale: A histogram shows the shape of the distribution. Here, the strong right skew motivates

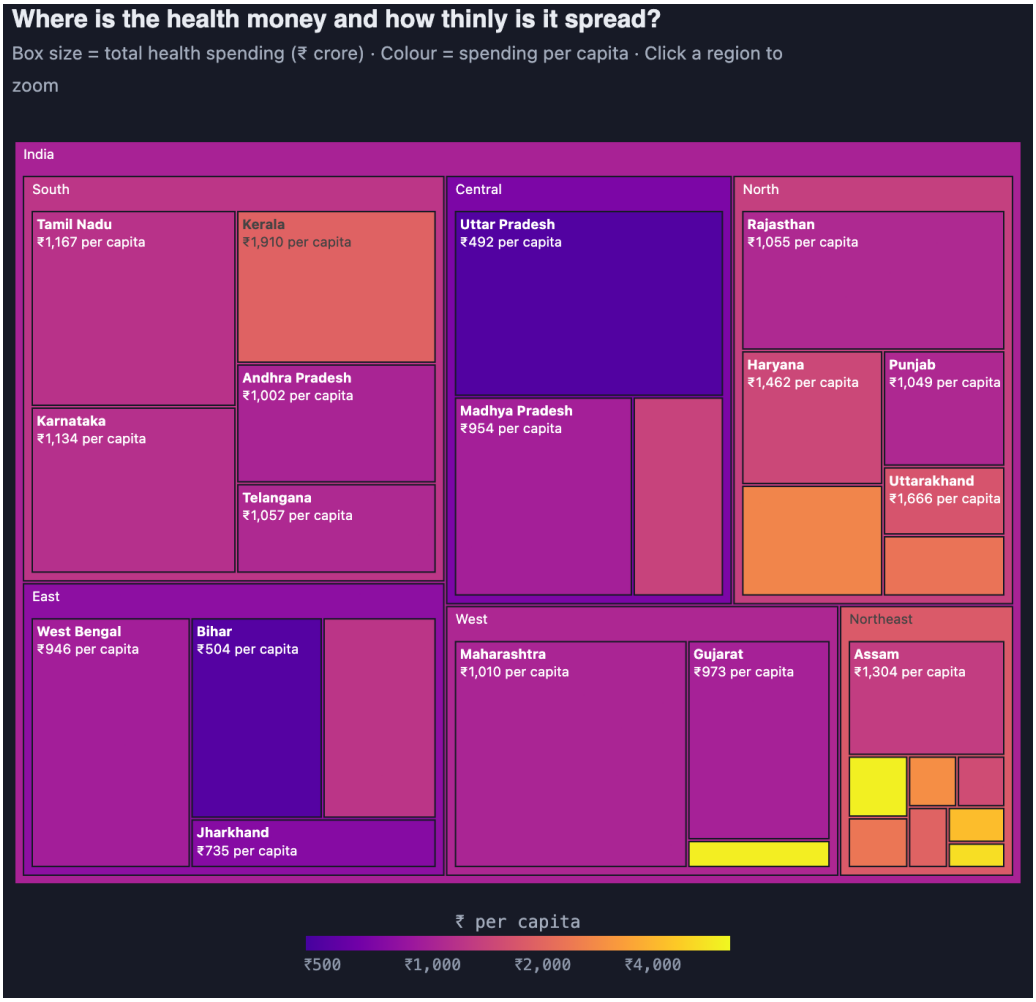


Figure 8: The five biggest budgets (Maharashtra, Uttar Pradesh, West Bengal, Tamil Nadu, Rajasthan) account for 38% of all state health spending but serve 46% of the population. The boxes for the north-eastern states are small but bright: together, this region spends Rs. 1,759 per capita, compared with Rs. 988 for all 29 states combined.

the use of logarithmic scales and rank-based statistics in the rest of the dashboard.

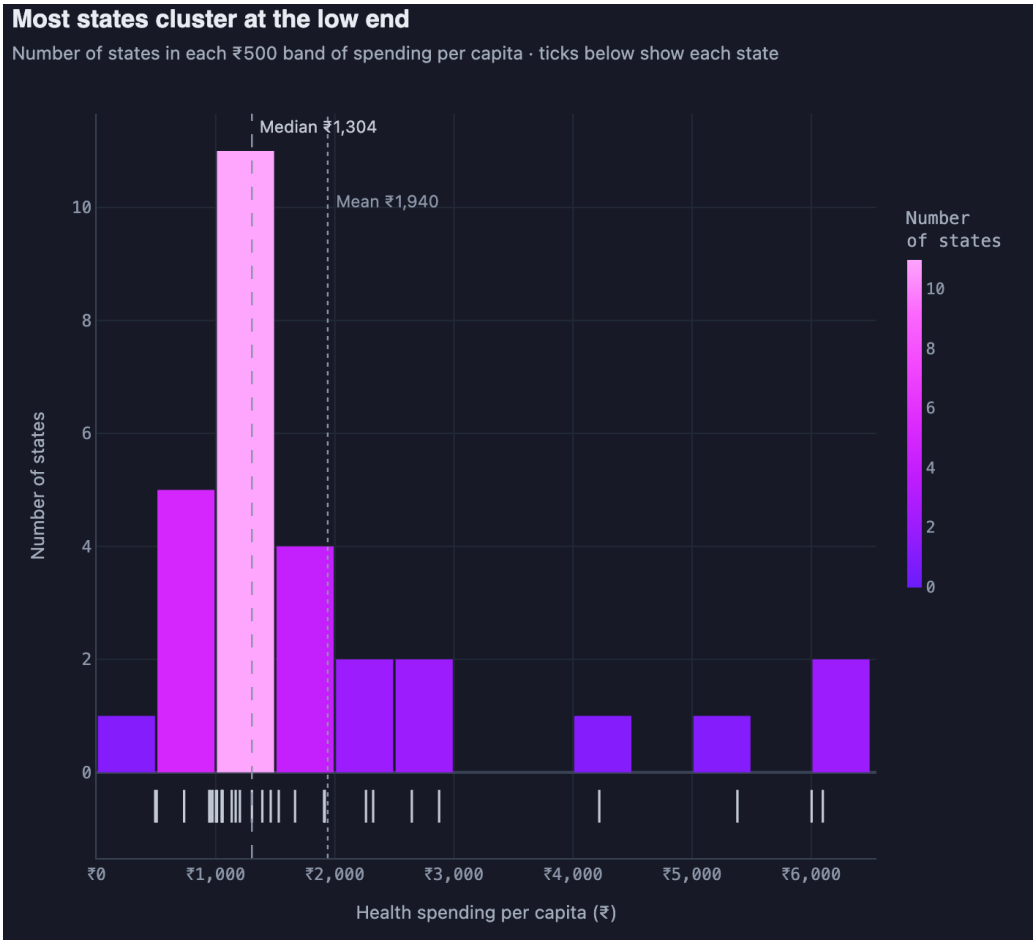


Figure 9: 17 of 29 states spend under Rs. 1,500 per capita, while 4 states spend over Rs. 4,000. The mean (Rs. 1,940) is considerably higher than the median (Rs. 1,304), reflecting the right-skewed distribution. This motivates the use of Spearman's rank correlation and logarithmic scales.

Most states spent less on health than they budgeted (dot plot, bivariate)

Chart Description: Budget utilisation: each state's actual spending and Revised Estimate as a percentage of its original Budget Estimate.

Interpretation: The vertical line at 100% represents spending exactly as budgeted. The filled orange dot represents actual spending (Accounts), while the hollow blue diamond represents the Revised Estimate. The grey bar shows the difference between actual spending and 100%. States are listed in the same order as in the spending ranking, allowing a row-by-row comparison with the lollipop chart.

Rationale: It shows how closely actual expenditure matches the original budget, supporting the decision to use actual spending (Accounts) throughout the analysis.

Summary of Tab 1

Per capita health spending varies twelve-fold across states and is concentrated at the lower end, with a long tail of small states. Many of the states that spend more per capita lie in the Northeast. Total budgets largely reflect state size and most states spent less than they had budgeted. For these reasons, the rest of the analysis compares spending per capita using actual expenditure, plots it on a logarithmic scale and relates it to outcomes using rank-based correlations. The six north-eastern states are also examined separately in Tab 4.

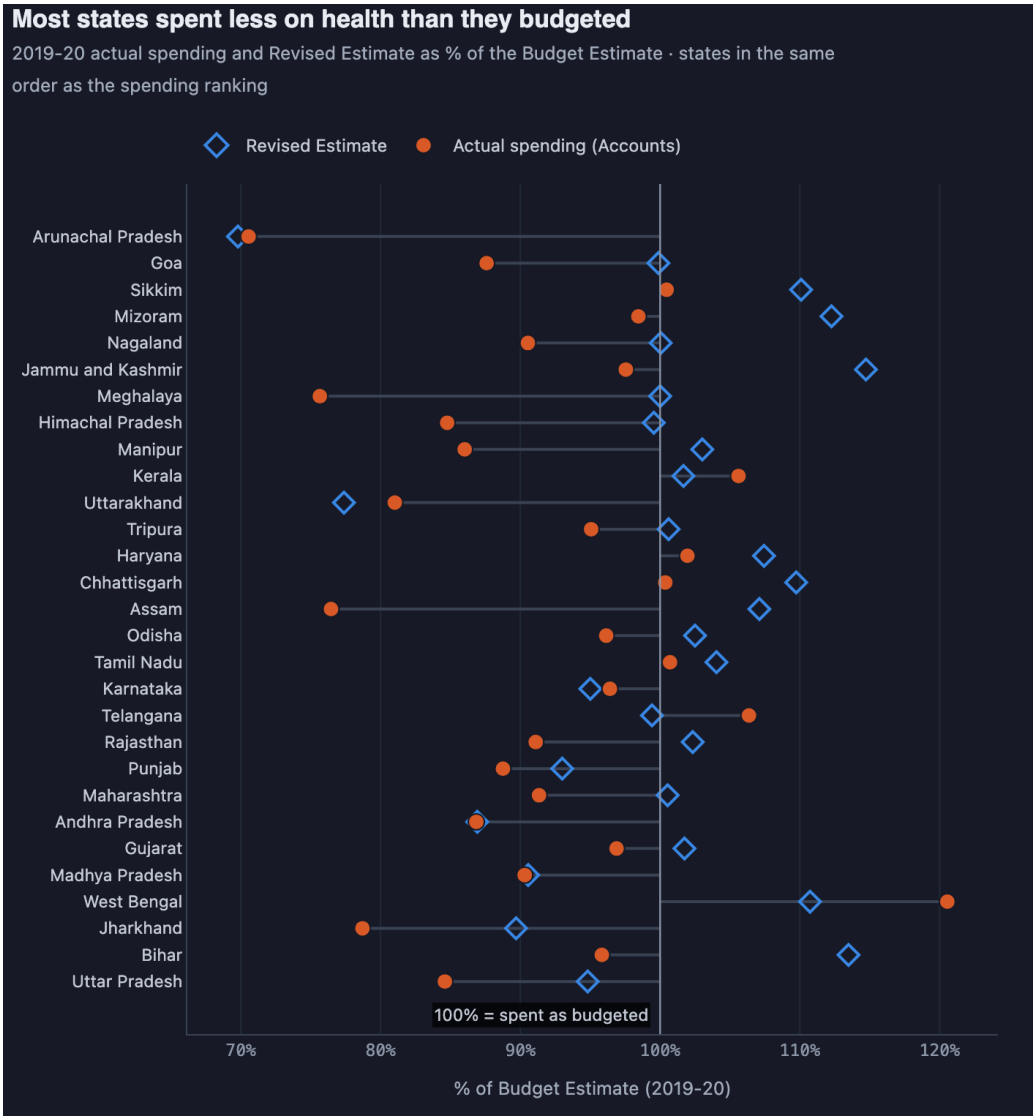


Figure 10: In 2019-20, 22 of 29 states spent less on health than they budgeted, by a median of 10%. Arunachal Pradesh spent only 71% of its Budget Estimate, while West Bengal spent 121%. This shows that the original budget estimates did not always reflect actual expenditure.

4.7 Tab 2: Health Outcomes

Tab 2 focuses on the health of mothers and children across states and the care that they receive. The four headline figures at the top of this tab are:

Headline	Value	Meaning
Median infant mortality	28	Deaths per 1,000 live births, ranging from 4 (Kerala) to 50 (Uttar Pradesh)
Median child stunting	32%	Children under 5 who are too short for their age
Median anaemia in pregnancy	46%	Pregnant women who are anaemic
Sex ratio at birth below normal	12 of 29	States with fewer than 935 girls born per 1,000 boys

Table 5: Headline figures for Tab 2.

Health outcomes and care received (heatmaps, multivariate)

Chart Description: Two heatmaps display the scores of every state across individual indicators. One heatmap covers the 10 outcome indicators, while the other one does the same for the 9 care indicators.

Interpretation: Each row represents a state and each column represents an indicator. States are arranged from highest to lowest per capita health spending, with the highest-spending state at the top. Each cell shows the indicator score calculated as per Section 4.2, where 0 represents the worst-performing state and 100 the best-performing state. Lighter shades always indicate better performance, regardless of whether a higher or lower value is desirable for the original indicator. The indicators are grouped into blocks: child survival, child nutrition and mothers' health for outcomes; and pregnancy and birth, and young children for care. Each block has its own colour scale and colour bar. If spending were positively associated with an indicator, the cells towards the top of its column would generally be lighter than those towards the bottom.

Rationale: The heatmaps allow all 29 states and all 19 indicators (551 values) to be compared in a single view. Ordering the states by per capita spending makes it easier to look for a pattern from top to bottom. Separate colour scales for each indicator block also help distinguish the groups.

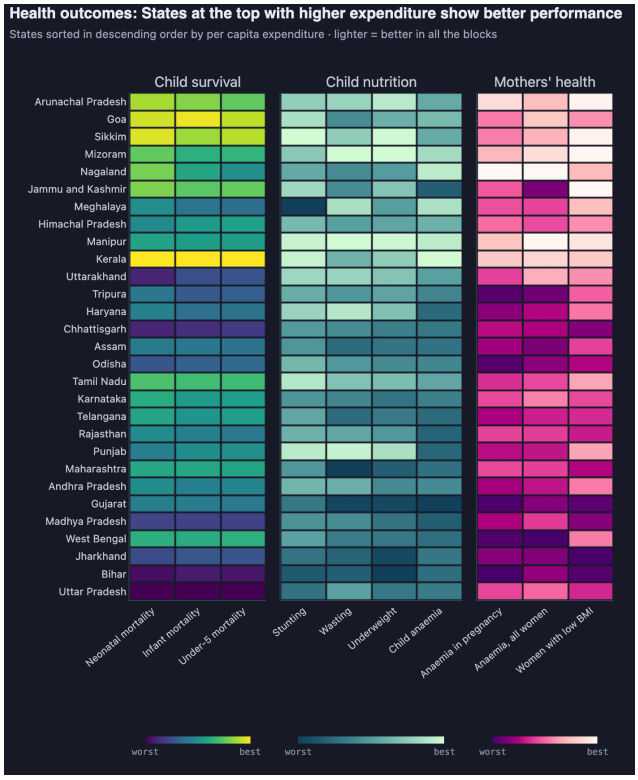
Together, the heatmaps provide an initial indication of the main finding: health outcomes tend to be better among higher-spending states, whereas care indicators show no comparable pattern. The averages by spending group use all the indicators displayed in each heatmap, while the correlations only use the corresponding composite indices.

More care does not always result in good outcomes (quadrant scatter plot, bivariate)

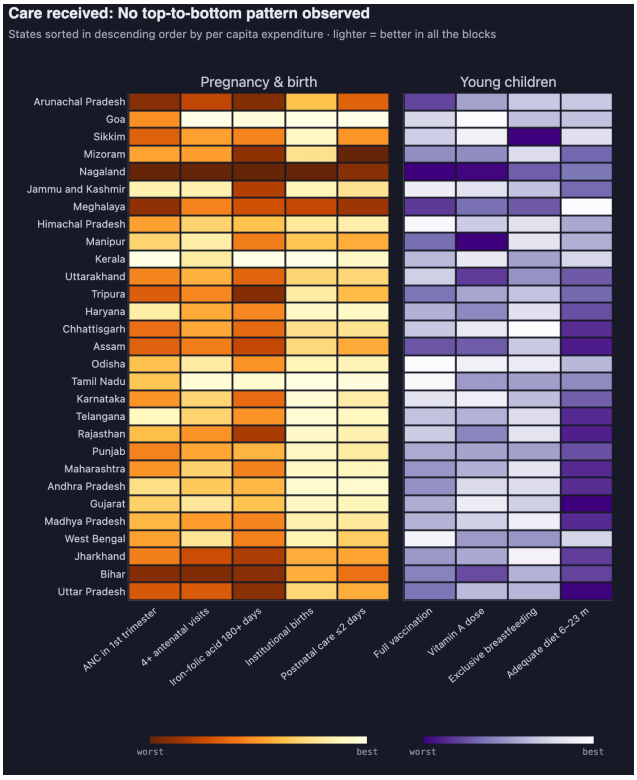
Chart Description: The scatter plot compares each state's Care Index on the horizontal axis with its Outcome Index on the vertical axis.

Interpretation: Each marker represents a state and is coloured and shaped according to its region. Dotted lines mark the median values of the two indices, dividing the chart into four quadrants. The quadrants are labelled to describe the combination of care and outcomes; for example, the top-left quadrant contains states with below-median care but above-median outcomes. Four states (Arunachal Pradesh, Nagaland, Gujarat and Odisha) are labelled: the two with the largest differences in each direction between their Outcome Index and Care Index.

Rationale: If care were the main link between spending and health outcomes, the two indices would be expected to increase together, with states lying closer to a diagonal pattern. The quadrant layout makes it easier to identify states that do not follow this pattern.



(a) Health outcomes. The highest-spending third of states averaged 70/100, compared with 31/100 for the lowest-spending third ($\rho = +0.77$, $p < 0.001$, $n = 29$).



(b) Care received. The highest-spending third averaged 50/100, compared with 54/100 for the lowest-spending third; the correlation is not significant ($\rho = -0.08$, $p = 0.68$, $n = 29$).

Figure 11: Indicator scores (0 = worst state, 100 = best) for every state, sorted by spending per capita, with the highest spender at the top.



Figure 12: Care and outcomes are only weakly related across states ($\rho = +0.14$, $p = 0.46$, $n = 29$). Seven states have better-than-median outcomes despite having below-median care; six of these are in the Northeast. This suggests that factors beyond the health system, such as diet, sanitation and women's schooling, may also influence health outcomes.

Nagaland (Care Index 1, Outcome Index 64) and Arunachal Pradesh (26 and 78) have the largest positive gaps between outcomes and care. On the other hand, Gujarat (73 and 17) and Odisha (77 and 37) have the largest negative gaps.

Rural vs urban (dumbbell chart, bivariate)

Chart Description: The chart compares the urban and rural values of an indicator for each state. Infant mortality is shown here, while the dashboard allows the selection of other indicators.

Interpretation: Each row represents a state, arranged by per capita spending. The blue circle represents the urban value and the orange diamond the rural value. A grey line connects the two markers, showing the size of the rural-urban gap. Hollow markers identify values imputed during data cleaning (Section 3), making it possible to distinguish between estimates and observed values.

Rationale: A dumbbell chart makes it easy to compare two values for each state and see the difference between them. Sorting states by spending also helps examine whether the rural-urban gap is smaller in higher-spending states, which could indicate whether additional spending is associated with better outcomes in rural areas.

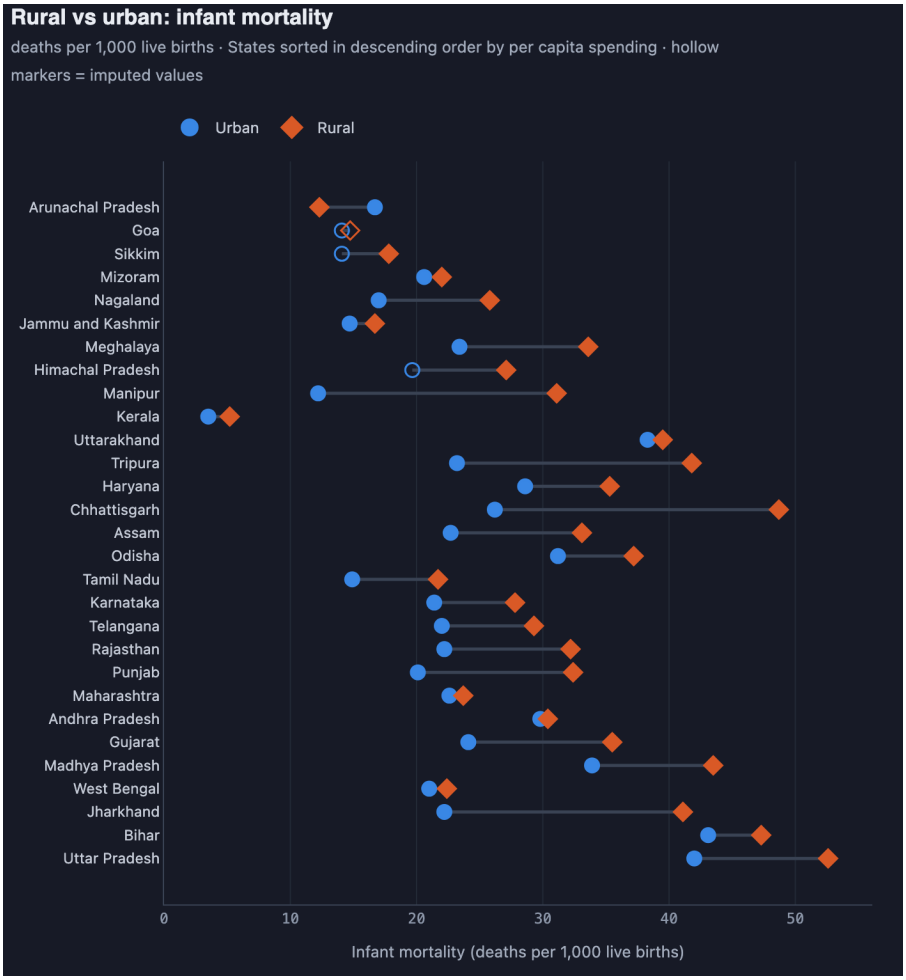


Figure 13: Rural areas have higher infant mortality than urban areas in 28 of 29 states, with a median gap of 6.8 deaths per 1,000 live births. Arunachal Pradesh is the exception. Higher spending does not clearly narrow the rural–urban gap ($\rho = +0.28$, $p = 0.15$, $n = 29$), so there is no clear evidence that additional spending is associated with a smaller gap in rural areas.

How far apart the states are on four headline outcomes (box plots, univariate)

Chart Description: The box plots show the distribution across states of four headline health outcomes: infant mortality, stunting, child anaemia and anaemia in pregnancy.

Interpretation: Each panel represents one outcome in its original unit. The box covers the middle 50% of states, from the 25th to the 75th percentile, and the line inside it marks the median. The whiskers extend to the most extreme values within 1.5 times the interquartile range. Each dot represents a state. The dots are spread horizontally to prevent overlapping and are coloured by region. Lower values indicate better outcomes for all four indicators.

Rationale: Box plots summarise the variation in each outcome, while the individual markers retain information about all 29 states. Showing the indicators in their original units, rather than as 0-100 scores, also makes the scale and severity of each health problem easier to comprehend.

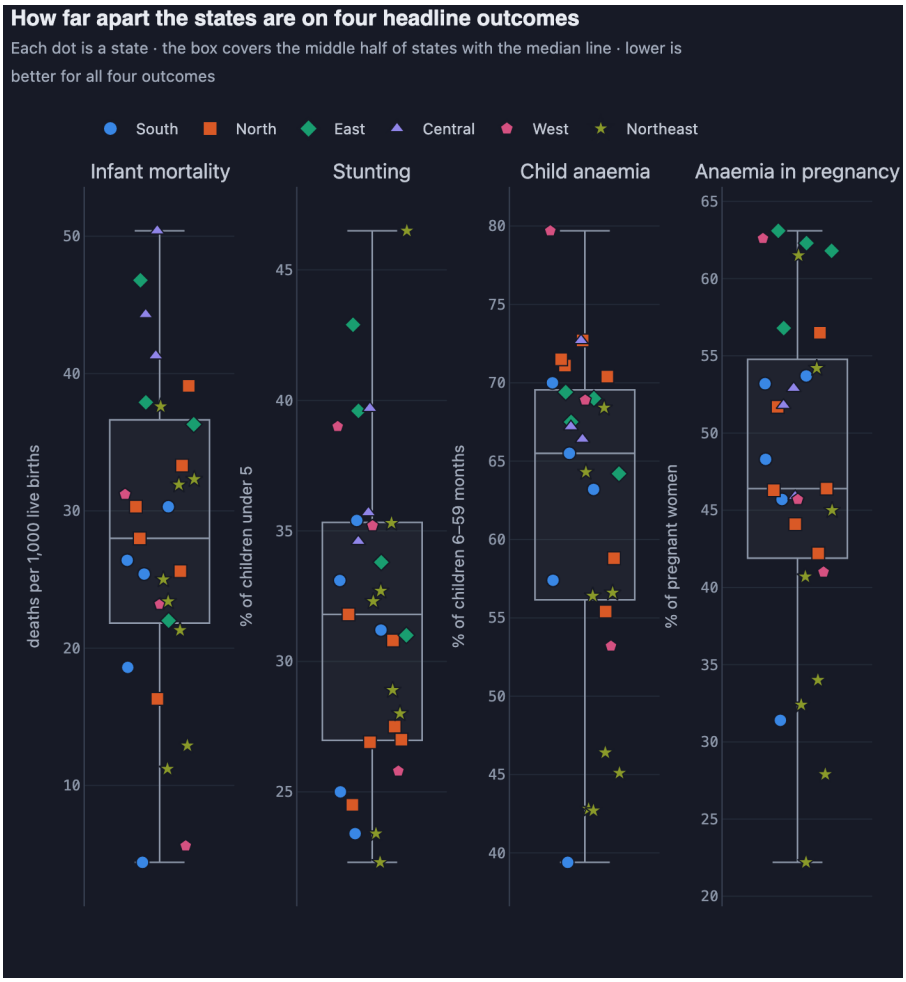


Figure 14: Infant mortality ranges from 4.4 deaths per 1,000 live births in Kerala to 50.4 in Uttar Pradesh, a difference of more than eleven times. Stunting ranges from 22.3% in Sikkim to 46.5% in Meghalaya. Anaemia, by contrast, is widespread: at least 66% of young children and 46% of pregnant women are anaemic in half of the states.

Many states have too few girls born (lollipop chart against a reference range, univariate)

Chart Description: The chart shows the sex ratio at birth for each state, measured as the number of girls born per 1,000 boys among children born in the five years preceding the NFHS-5 survey.

Interpretation: In the absence of sex selection, the expected range is approximately 103 to 107 boys for every 100 girls, equivalent to around 935 to 971 girls per 1,000 boys. This natural range is shaded, and the dashed line marks the expected value of 952. Each state's marker is connected to this reference value by a grey line, with the length indicating the difference. Marker colour ranges from white (no shortfall) to deep red (880 or fewer girls per 1,000 boys), highlighting states with the largest shortfalls.

Rationale: Comparing each state with a reference range shows not only how states rank against one another but also whether their sex ratios fall outside the expected biological range. The red colour

scale draws attention to the states with the largest shortfalls.

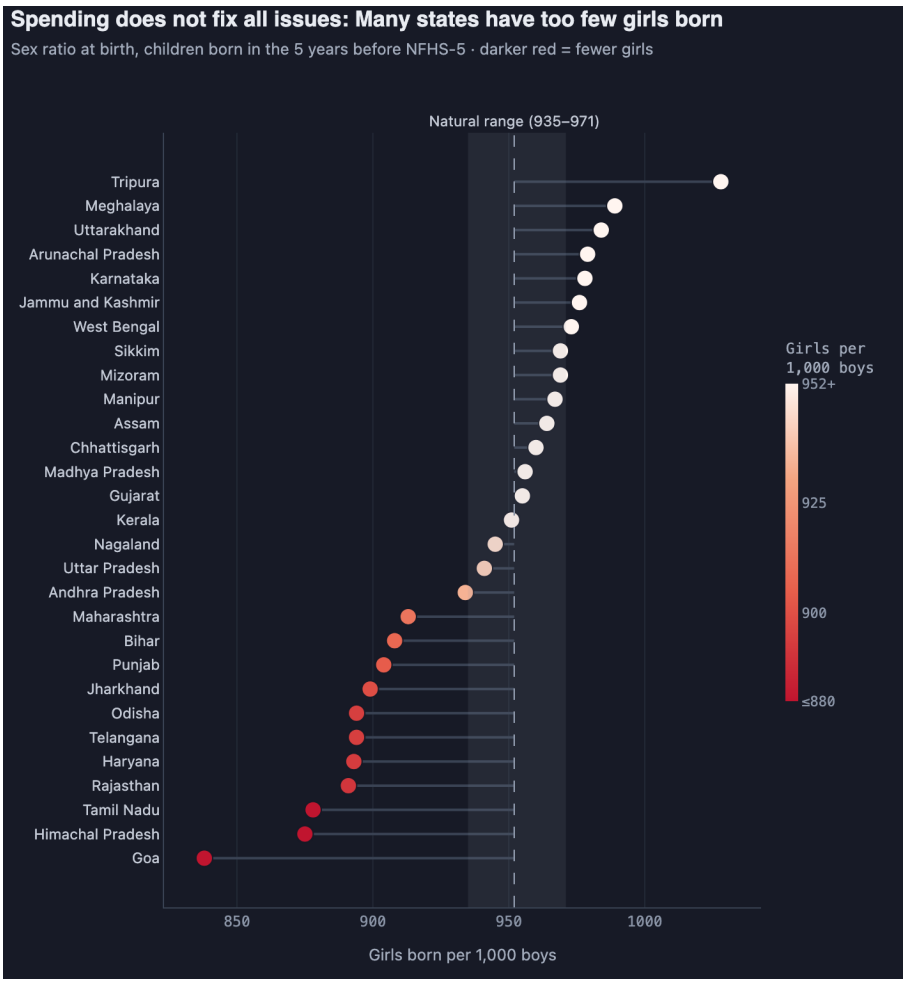


Figure 15: Twelve of the 29 states have sex ratios below the natural range, with the lowest recorded in Goa (838 girls per 1,000 boys). This pattern is consistent with sex-selective abortion. The shortfall is not explained by health spending ($\rho = +0.27$, $p = 0.16$, $n = 29$) or women's schooling ($\rho = -0.24$, $p = 0.22$, $n = 29$). Some of the best-educated states, including Goa and Himachal Pradesh, have the lowest ratios. This may point to son preference alongside access to sex-determination technology. Estimates for small states should be interpreted with caution.

The sex ratio at birth should be interpreted cautiously for small states because the NFHS-5 estimate is based on a sample of recent births, which may be relatively small in these states. For Goa, the Civil Registration System, which records almost all births in the state, reports 947 girls per 1,000 boys in 2020, compared to 838 in NFHS-5. A similar phenomenon is also observed in Himachal Pradesh: 936 in the Civil Registration System versus 875 in NFHS-5.

Summary of Tab 2

Health outcomes vary substantially between states: infant mortality differs by more than 11 times, while stunting varies by roughly two times. Higher-spending states tend to have better health outcomes, but care indicators do not show a comparable relationship with spending. Care and health outcomes are also only weakly related. Rural areas have higher infant mortality than urban areas in almost all states, and anaemia remains widespread. The sex ratio at birth falls below the natural range in 12 states, with no statistically significant association with health spending. These descriptive patterns are tested formally in Tabs 3 and 4.

4.8 Tab 3: Spending vs Outcomes

Tab 3 addresses the main question directly: do states that spend more on health per capita achieve better maternal and child health outcomes? It examines the relationship between per capita health spending (Tab 1) and the Outcome Index and Care Index (Tab 2), first at the index level and then across individual indicators. The four headline figures at the top of this tab are:

Headline	Value	Meaning
Spending vs Outcome Index	$\rho = +0.77$	Strong positive association ($p < 0.001$)
Spending vs Care Index	$\rho = -0.08$	No statistically significant association ($p = 0.68$)
Outcomes linked to spending	10 of 10	Every health outcome is significantly better in higher-spending states
Furthest above the prediction	Kerala	33 points above the Outcome Index predicted by its spending

Table 6: Headline figures for Tab 3.

All correlations in this tab are Spearman's ρ, calculated between per capita spending and the measure being examined across the 29 states. The corresponding p-values and 95% confidence intervals are calculated as described in Section 4.3. A result is considered to be statistically significant when $p < 0.05$.

Health spending per capita vs Outcome Index (bubble scatter plot, bivariate)

Chart Description: This is the main chart of the dashboard. It plots each state's health spending per capita on the horizontal axis against its Outcome Index on the vertical axis.

Interpretation: Each bubble represents a state, with its colour and shape indicating the region. The *area* of each bubble is proportional to the state's population, making the difference between a large state such as Uttar Pradesh (about 230 million people) and a small state such as Sikkim (under one million) visible. The horizontal axis uses a logarithmic scale where equal distances represent equal ratios of spending. The dashed line represents the trend line fitted across all 29 states:

$$\widehat{\text{Outcome Index}} \;=\; a + b\,\log_{10}(\text{spending per capita}), \qquad a = -132.6, \quad b = 57.0,$$

where a and b are estimated using ordinary least squares. The slope indicates that a tenfold increase in spending is associated with an expected increase of approximately 57 points in the Outcome Index. A doubling of spending corresponds to approximately $57.0 \times \log_{10} 2 \approx 17$ points. Hollow bubbles indicate states whose values were partly imputed during data cleaning (Section 3). The indicator displayed on the vertical axis can be changed. It is also possible to switch between Total, Urban and Rural figures and exclude the six north-eastern states.

Rationale: A scatter plot provides a direct way to examine the relationship between two numerical variables. The logarithmic scale is appropriate because spending is strongly right-skewed (Tab 1). On a linear scale, the four highest-spending states would stretch the horizontal axis, compressing most other states into a small region. Sizing bubbles by population also emphasises that each observation represents a state, not an individual: a small state and a large state each contribute one observation to the correlation. The option to exclude the six north-eastern states provides a robustness check, as these states have relatively high per capita spending and distinctive outcomes.

The trend line is used to describe the overall pattern and calculate the prediction gaps examined later in this tab. It should not be interpreted as a causal model. The strength of the association is measured using Spearman's ρ, which is unaffected by the logarithmic transformation because taking logarithms does not change the ranks.



Figure 16: States that spend more per capita have higher (better) Outcome Index scores ($\rho = +0.77$, $p < 0.001$, $n = 29$). The median Outcome Index is 27.4 in the lowest-spending quarter of states (Rs. 492–1,010 per capita) and 70.6 in the highest-spending quarter (Rs. 2,322–6,097). Kerala scores highest (87.2) while spending Rs. 1,910 per capita, whereas Bihar scores lowest (10.2) at Rs. 504. Excluding the six north-eastern states, the correlation remains positive and significant ($\rho = +0.68$, $p < 0.001$, $n = 23$).

Spending tracks health outcomes, not the care received (dot plot with confidence intervals, bivariate)

Chart Description: The chart displays the Spearman correlation between per capita spending and each individual indicator, together with its 95% confidence interval.

Interpretation: Each row represents an indicator, grouped into three panels: health outcomes, care received, and cost and cover for families. The dot represents ρ, while the horizontal bar shows its 95% confidence interval. The correlations are *signed* so that a positive value always indicates an association between higher spending and better results. For indicators where lower values are better, such as infant mortality and stunting, the sign of the correlation is flipped. For example, the correlation with infant mortality is $\rho = -0.57$, which is displayed as $+0.57$. Filled dots indicate statistically significant correlations ($p < 0.05$), while hollow dots indicate non-significant results. If a confidence interval crosses the vertical zero line, the data are consistent with no association.

Rationale: Examining each indicator separately helps determine whether the overall index result is reflected across individual measures or driven by only a few indicators. Signing the correlations places all indicators in the same direction, making their results easier to compare. Confidence intervals also show the uncertainty in each estimate, which is important with a sample of only 29 states.

The out-of-pocket expenditure result (unsigned $\rho = +0.70$) is examined further in Tab 4.

The top spenders lead on every outcome group, but not on care (grouped bar chart, bivariate)

Chart Description: The chart compares the average scores of the lowest-spending and highest-spending quarters of states across each group of indicators.

Interpretation: The 29 states are divided into four groups (quarters) according to per capita spending, but only the two extreme quarters are compared. For each indicator group, the score is calculated as the average of the 0-100 indicator scores across all indicators in that group and all states in the quarter. The orange bars represent the lowest-spending quarter, while the blue bars represent the highest-spending quarter. The three groups above the dotted line correspond to health outcomes (child survival, child nutrition and mothers' health), while the two below it represent care (pregnancy and birth, and young children).

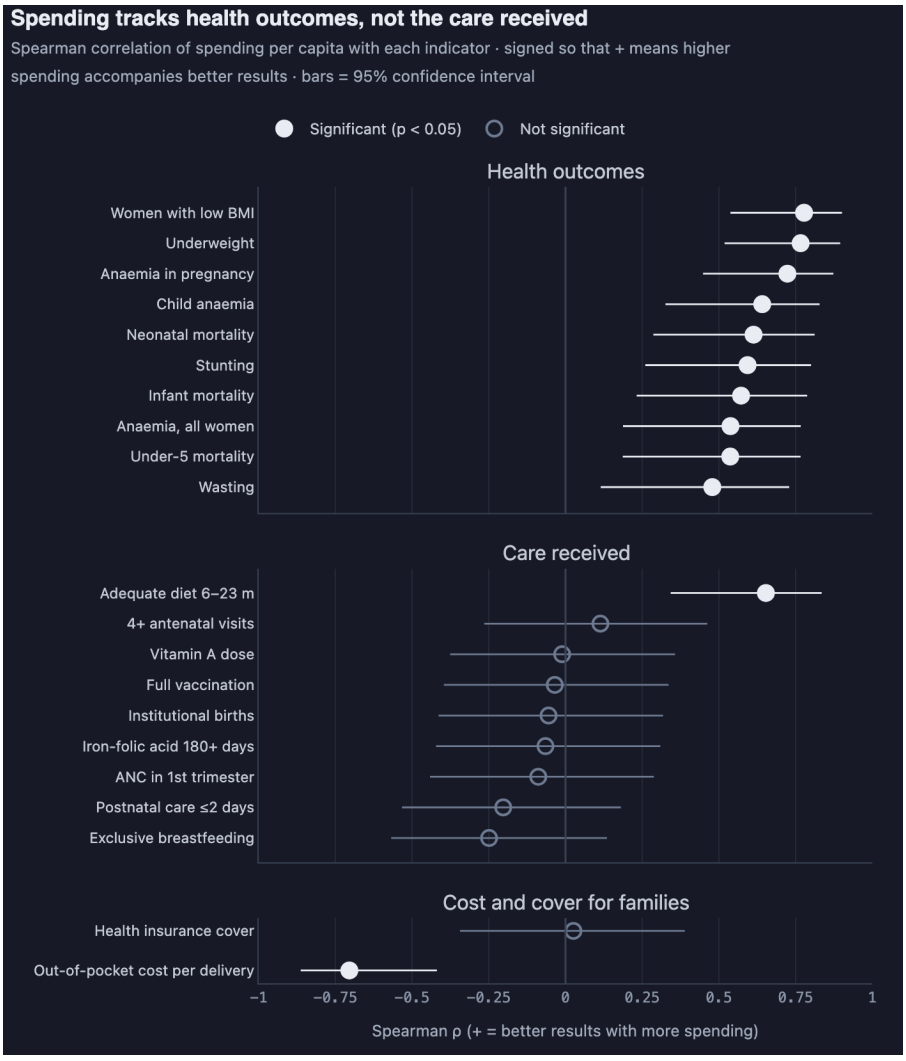


Figure 17: All 10 health outcomes are significantly better in states that spend more, with the strongest association for women with low BMI. Only 1 of the 9 care indicators shows a significant positive association with spending (adequate diet for children aged 6–23 months). One indicator moves in the opposite direction: families pay more out of pocket for delivery in higher-spending states.

Rationale: Correlation coefficients can be difficult to interpret on their own. Comparing the lowest- and highest-spending quarters translates the relationship into a more direct comparison of performance on the same 0-100 scale. Horizontal bars keep the group labels readable, while placing the two spending groups side by side makes their differences easier to compare.

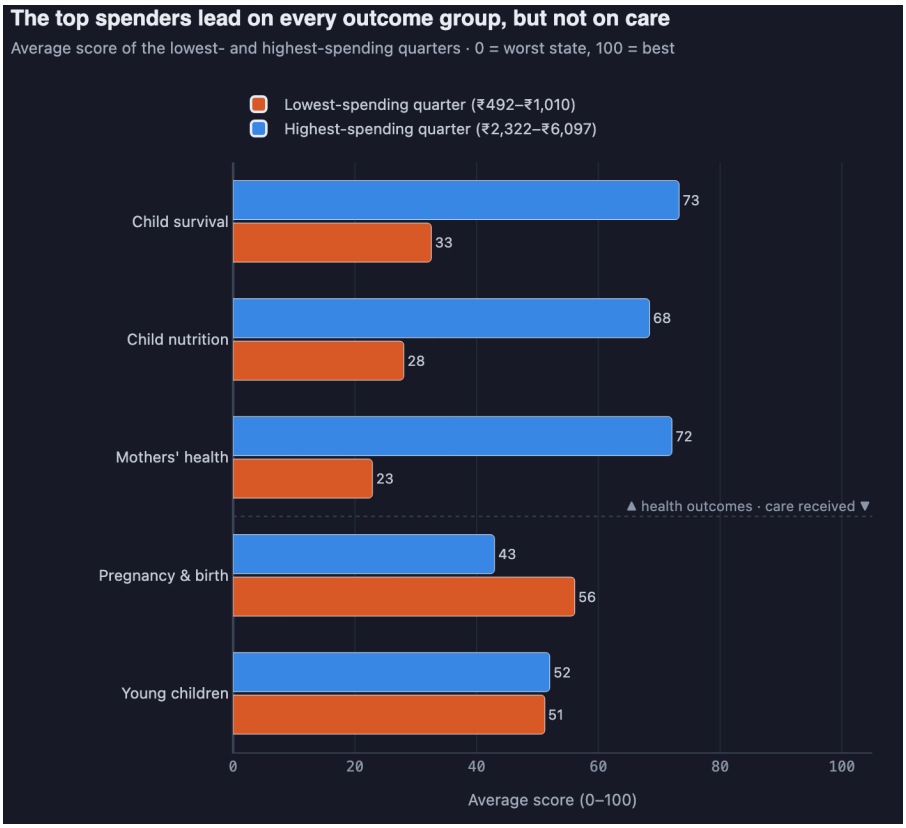


Figure 18: The highest-spending quarter of states scores substantially higher across every outcome group. For mothers' health, the average score is 72 out of 100, compared with 23 in the lowest-spending quarter. However, the highest-spending quarter does not lead on care: it scores lower on care during pregnancy and birth (43 against 56) and almost the same on care for young children (52 against 51).

Who does better or worse than their spending predicts (diverging bar chart, univariate)

Chart Description: The chart shows the difference between each state's actual Outcome Index and the value predicted by its spending using the trend line from the main scatter plot.

Interpretation: The gap for each state is calculated as

$$\begin{aligned} \text{gap} &= \text{Outcome Index} - \widehat{\text{Outcome Index}} \\ &= \text{Outcome Index} - (a + b \log_{10}(\text{spending per capita})), \end{aligned}$$

which represents the vertical distance between the state's bubble and the dashed trend line. A positive gap (blue bar, extending to the right) indicates that a state performs better than predicted by its spending. A negative gap (orange bar, extending to the left) means that it performs worse than predicted. States are arranged from the largest positive gap at the top to the largest negative gap at the bottom. For example, for Kerala:

$$\begin{aligned} \widehat{\text{Outcome Index}} &= -132.6 + 57.0 \times \log_{10}(1{,}910) = -132.6 + 57.0 \times 3.281 \approx 54.3, \\ \text{gap} &= 87.2 - 54.3 \approx +33. \end{aligned}$$

Rationale: Spending accounts for much of the variation in outcomes across states, but it does not explain all of it. The prediction gaps make the remaining differences visible and identify states that

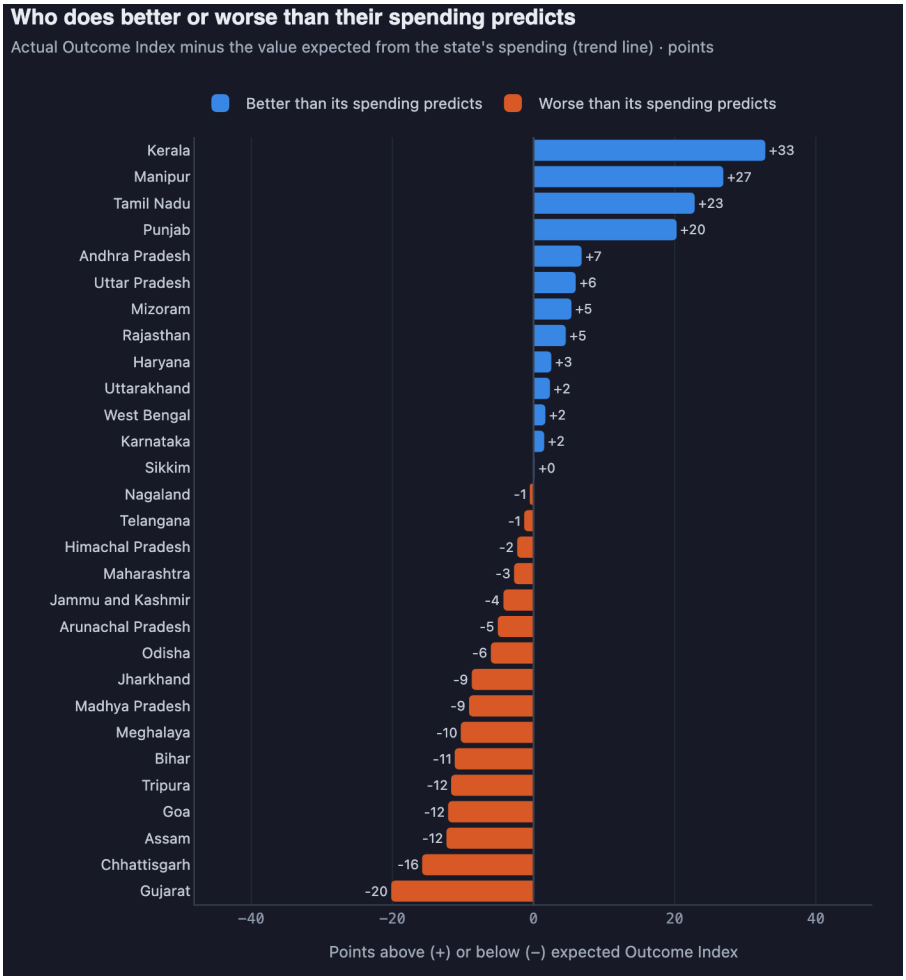


Figure 19: Kerala scores 33 points above the Outcome Index predicted by its spending of Rs. 1,910 per capita, the largest positive gap among the states. Gujarat scores 20 points below its prediction at Rs. 973 per capita. Overall, 13 of the 29 states perform better than predicted, showing that spending alone does not determine outcomes.

perform notably better or worse than expected. A diverging bar chart centred on zero makes both the direction and magnitude of each gap easy to compare.

As noted in Section 4.3, the prediction gap is descriptive: a positive gap does not necessarily mean that a state uses its money more effectively. Kerala's outcomes, for example, may also reflect factors beyond its 2019-20 health budget, including long-term investments in women's education and primary care. Some of these factors are examined in Tab 4.

Outcomes rise with spending, care does not (side-by-side scatter plots, bivariate)

Chart Description: Two scatter plots display the same 29 states side by side. The left panel plots per capita spending against the Outcome Index, while the right panel plots spending against the Care Index.

Interpretation: Both panels use the same logarithmic horizontal axis for spending and the same 0-100 vertical scale for the indices, allowing direct comparison. Markers are coloured and shaped according to region. The trend line is solid when the correlation is statistically significant and dotted when it is not.

Rationale: Placing the two indices side by side with identical axes immediately makes their contrasting relationships with spending visible. The Outcome Index shows a clear upward pattern, while the Care Index does not.

Summary of Tab 3

States that spend more on health per capita tend to have substantially better maternal and child health outcomes. The correlation with the Outcome Index is strong ($\rho = +0.77$), all 10 individual health outcomes are significantly better in higher-spending states, and the highest-spending quarter performs better than the lowest-spending quarter across all outcome groups. The relationship with care received is much weaker: the Care Index does not show a significant association with spending across all 29 states ($\rho = -0.08$), and only 1 of the 9 care indicators improves significantly with spending. However, care shows a positive association once the six north-eastern states are excluded ($\rho = +0.44$). Some states, particularly Kerala, perform substantially better than their spending predictions, while others, such as Gujarat, perform worse. These findings describe associations rather than causal effects. Tab 4 examines whether the relationship between spending and outcomes remains after accounting for other factors, unusual states and the timing of the survey.

4.9 Tab 4: Is It Really the Spending?

Tab 3 showed a strong association between health expenditure per capita and maternal and child health outcomes. However, an association alone does not establish that spending is responsible for better outcomes. Tab 4 therefore examines three alternative explanations:

1. **Confounding:** States that spend more may also have advantages in other areas, such as women's education or sanitation. These factors, rather than spending itself, could explain their better outcomes.
2. **Unusual states:** The six north-eastern states have the highest per capita spending and were flagged as outliers in Section 3. They might therefore be driving the overall result.
3. **Timing of the survey:** NFHS-5 was conducted in two phases. The second phase took place during the COVID-19 pandemic, when health services were disrupted.

The tab also includes a fourth check that examines what families pay out of pocket for healthcare. The four headline figures at the top of this tab are:



Figure 20: Left: higher-spending states have clearly better maternal and child health outcomes ($\rho = +0.77$, $p < 0.001$). Right: the amount of antenatal care, facility births and vaccination received is not significantly associated with spending ($\rho = -0.08$, $p = 0.68$). This pattern is partly influenced by the six north-eastern states, which spend more but have relatively low levels of formal care. Excluding these states, care shows a positive association with spending ($\rho = +0.44$, $n = 23$).

Headline	Value	Meaning
Lowest link after adjusting	$\rho = +0.52$	After allowing for women's literacy (from +0.77)
Without the north-east	$\rho = +0.68$	23 states ($p < 0.001$)
Surveyed before COVID	$\rho = +0.76$	18 Phase 1 states ($p < 0.001$)
Out-of-pocket cost vs spending	$\rho = +0.70$	Families pay more per delivery where states spend more

Table 7: Headline figures for Tab 4.

How the confounding check works

A factor z may help explain the association between spending and outcomes if it is related to both. In other words, states with higher spending must tend to have more of z, and states with more of z must also tend to have better outcomes. To examine this, the partial Spearman correlation $\rho_{xy \cdot z}$ (Section 4.3) is calculated between spending (x) and the Outcome Index (y), after removing the part of each variable associated with z. In effect, this compares states with similar values of z. The result is interpreted as follows:

- if $\rho_{xy \cdot z}$ remains close to the original $\rho = +0.77$, the factor does not explain much of the association;
- if it falls substantially but remains significant, the factor may explain part of the association;
- if it falls to around zero, the factor may account for most or all of the association.

The share of the original association that remains is calculated as

$$\text{Retained} \;=\; \frac{\rho_{xy \cdot z}}{\rho_{xy}} \times 100\%.$$

Five factors were selected from the NFHS-5 control variables collected in Section 2: women with 10 or more years of schooling, women's literacy, improved sanitation, improved drinking water and health insurance coverage. Each was tested separately because, with only 29 states, adjusting for several factors simultaneously would leave too few degrees of freedom for a reliable estimate.

Is it the money, or women's schooling? (scatter plot with colour encoding, multivariate)

Chart Description: The chart plots health spending per capita on the horizontal axis (logarithmic scale) against the Outcome Index on the vertical axis. Each state's dot is coloured according to the share of women with 10 or more years of schooling.

Interpretation: The position of each dot is unchanged from the main scatter plot in Tab 3. The lighter dots represent those states where a larger share of women have completed 10 or more years of schooling. If schooling, rather than spending, explained the pattern, lighter dots would tend to appear towards the top right and darker dots towards the bottom left. Schooling can be replaced with any of the other four factors using the dropdown in the dashboard.

Rationale: Colour represents a third variable without requiring another axis. The relationship between spending and outcomes alongside a possible confounding factor can thus be examined in a single chart. Women's schooling is selected by default because it is often linked to maternal and child health.

For women's schooling, for example, Retained $= 0.72/0.77 \times 100\% \approx 93\%$.

The spending link survives every other explanation tested (dumbbell chart, bivariate)

Chart Description: The chart compares the correlation between spending per capita and the Outcome Index before and after adjusting for each of the five factors.

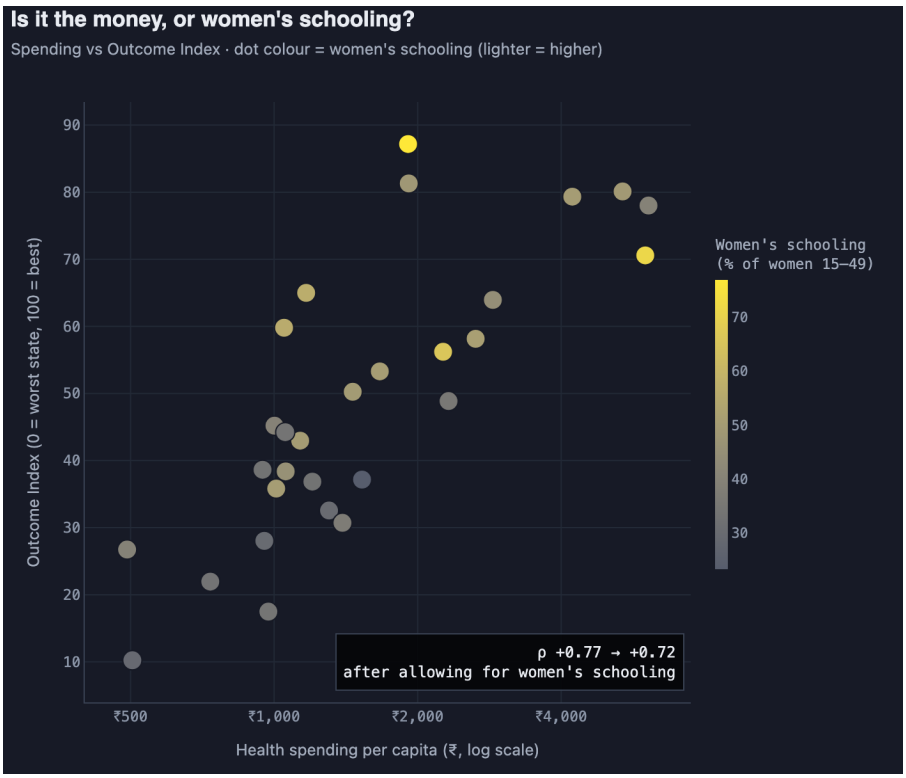


Figure 21: Higher-spending states also tend to have more women with 10 or more years of schooling ($\rho = +0.45$), which could partly explain their better outcomes. However, among states with similar levels of women's schooling, spending remains strongly associated with outcomes: the correlation falls only from +0.77 to +0.72 ($p < 0.001$), retaining about 93% of its strength.

Interpretation: Each row represents one factor. The grey dot shows the original correlation ($\rho = +0.77$), which is the same for every row, while the blue diamond shows the partial correlation after adjusting for that particular factor. The adjusted value is printed beside the diamond. The line joining the two markers shows how much the correlation changes: a longer line extending to the left indicates a larger reduction. The rows are sorted by the adjusted correlation, with the factor that reduces the correlation the most (women's literacy) at the bottom.

Rationale: A dumbbell chart displays the original and adjusted correlations together, making it easier to compare the extent of the changes across all five factors.

Factor (z)	**ρ with spending**	**ρ with Outcome Index**	**Partial ρ**	p
Women who are literate	+0.72	+0.73	+0.52	0.004
Improved sanitation	+0.69	+0.69	+0.57	0.001
Women with 10+ years of schooling	+0.45	+0.69	+0.72	< 0.001
Health insurance cover	+0.03	−0.04	+0.78	< 0.001
Improved drinking water	−0.36	−0.09	+0.80	< 0.001

Table 8: The five factors tested. The partial ρ is the correlation between spending per capita and the Outcome Index after adjusting for each factor ($n = 29$; original $\rho = +0.77$).

Table 8 helps explain the pattern in the chart. Women's literacy and improved sanitation are strongly associated with both spending and outcomes. Adjusting for these factors therefore reduces the correlation the most, although it still remains statistically significant in both cases. Health insurance has almost no association with spending, while improved drinking water is negatively associated with it. Neither of them displays the positive relationship with spending that would be expected of a fac-

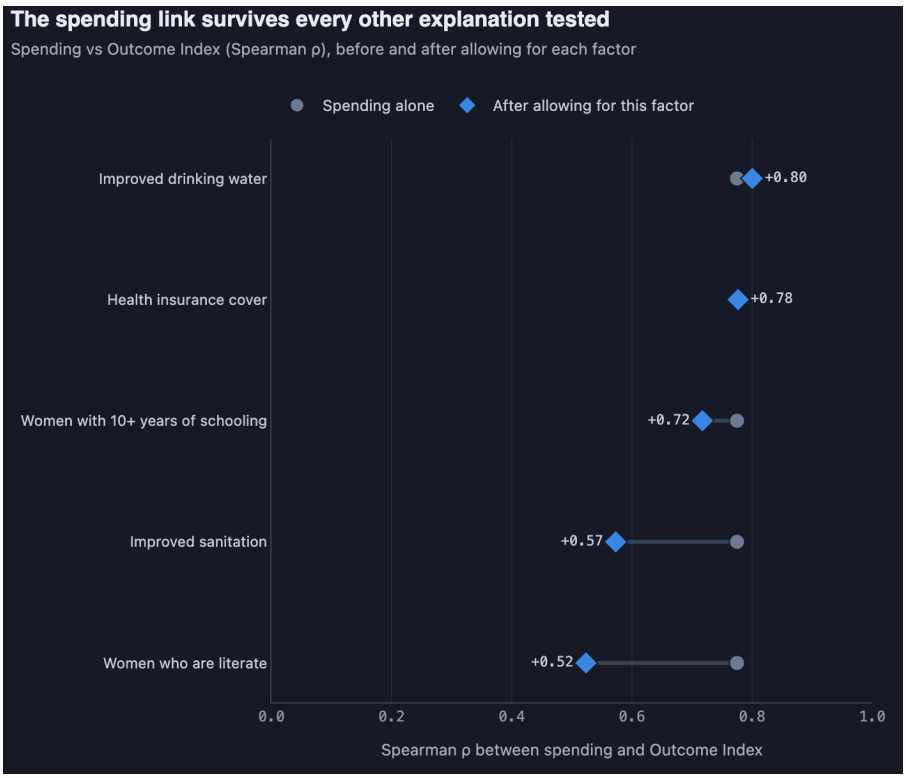


Figure 22: The association between spending and outcomes remains statistically significant after adjusting for each of the five factors. The correlation of +0.77 ranges from +0.52 to +0.80 after adjustment, with the largest reduction occurring when women's literacy is taken into account.

tor explaining the positive spending-outcome association. Adjusting for them leaves the correlation largely unchanged. Women's literacy and spending are themselves closely related ($\rho = +0.72$). With only 29 states, their separate contributions cannot be fully disentangled. The remaining correlation of +0.52 is the association that remains after accounting for literacy.

Do unusual states or COVID change the result? (dot plot comparing three groups, bivariate)

Chart Description: The chart shows the signed Spearman correlation between spending per capita and ten key measures, recalculated for three groups: all 29 states, the 23 states excluding the six north-eastern states flagged as outliers, and the 18 states that were surveyed before COVID-19 during Phase 1 of NFHS-5.

Interpretation: Each row represents one measure: the Outcome Index and the Care Index, followed by five individual outcomes (infant mortality, under-five mortality, stunting, child anaemia and anaemia in pregnancy), two care indicators (institutional births and full vaccination), and the out-of-pocket cost of a delivery. The correlations are once again signed so that a positive value indicates an association between higher spending and better results. The blue circle represents all 29 states, the orange diamond represents the states excluding the north-east and the hollow white square represents the Phase 1 states. The grey line spans the three values. A short line indicates that the result changes little between groups, while a long line indicates greater sensitivity to which states are included.

NFHS-5 Phase 1 fieldwork in the 18 states took place between June 2019 and January 2020, before the onset of the pandemic. The remaining 11 states were surveyed in Phase 2, largely during the pandemic, when routine services such as antenatal care and immunisation were disrupted. If the pandemic had substantially affected the Phase 2 data, the result for the Phase 1 states alone would be expected to differ from that for all 29 states.

Rationale: Showing the three groups on the same row makes the robustness check easy to compare visually. A result that remains similar across groups has the three markers close together. Different

marker shapes, as well as colours, help distinguish the groups.

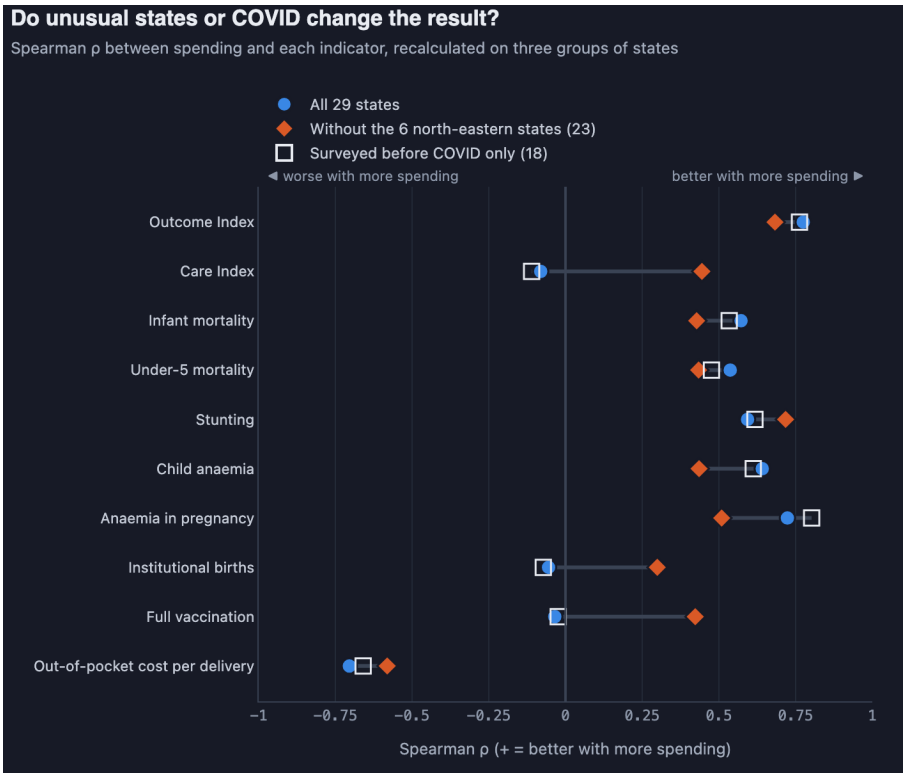


Figure 23: The main result is not driven by a few unusual states or by the pandemic: higher spending is associated with healthier mothers and children in all three groups ($\rho = +0.77$ for all states, $+0.68$ without the north-east and $+0.76$ for the 18 states surveyed before COVID-19). Care shows a different pattern: after excluding the north-east, the correlation changes from -0.08 to $+0.44$, indicating a positive association between spending and care. This may be because the north-eastern states have high spending but relatively low use of hospitals and clinics.

The two subsets overlap: five of the six north-eastern states (all except Arunachal Pradesh) were surveyed during Phase 1. Excluding the states surveyed during COVID-19 therefore does not remove the north-east, and the two checks examine different potential explanations.

Families pay more per delivery where states spend more (scatter plot, bivariate)

Chart Description: The chart plots the average out-of-pocket cost to a family for a delivery in a public health facility on the vertical axis (in rupees) against the state's health spending per capita on the horizontal axis (logarithmic scale).

Interpretation: Each marker represents a state and is coloured and shaped according to its region. The solid line is the trend line, fitted in the same way as in Tab 3: $\hat{y} = a + b \log_{10}(\text{pc})$. It is drawn as a solid line because the correlation is statistically significant. The three states with the highest costs and the state with the lowest cost are labelled.

Rationale: If higher public spending reduced the financial burden on households, out-of-pocket costs would be expected to fall as spending increased. This chart examines that expectation directly using the financial protection indicator collected for this purpose (Section 2).

This result does not mean that spending causes higher costs. Both may be associated with a third factor, such as differences in prices, transportation costs or incomes across states, which this data set does not measure.

Summary of Tab 4

The association between spending and health outcomes remains after all the checks in this tab. It is statistically significant after adjusting for each of the five factors tested (with partial correlations



Figure 24: One might expect families to pay less where the state spends more, but the opposite association is observed ($\rho = +0.70$, $p < 0.001$): a delivery in a public facility costs families Rs. 1,619 in Madhya Pradesh, compared with Rs. 14,518 in Manipur. The association remains when the north-eastern states are excluded ($\rho = +0.58$). One possible explanation is that care costs more in remote and hilly states, although the available data cannot confirm this.

ranging from +0.52 to +0.80), when the six north-eastern states are excluded (+0.68), and for the states surveyed before COVID-19 (+0.76). Women's literacy and sanitation account for part of the association, but not all of it. The apparent lack of an association between spending and care is largely influenced by the north-eastern states. Higher public spending also does not appear to reduce what families pay for a delivery. These checks strengthen the evidence for an association, but cannot establish causation: other factors that were not measured may still play a role, as discussed in the limitations (Tab 6).

4.10 Tab 5: Data Quality

Tab 5 presents the data-cleaning process described in Section 3 in a visual format. It describes the problems in the raw data, what was changed and how the data was checked before analysis. All the charts in this tab reproduce a step from the cleaning notebook (`data_cleaning.ipynb`), using the same methods and settings. No values are re-estimated here. The four headline figures at the top of this tab are:

Headline	Value	Meaning
Missing values filled	72 → 0	Filled by correlation-guided KNN imputation
Empty columns dropped	18	Columns with no data in the raw file
Text cells converted	83	Numbers stored as text with Indian-style commas
Outlier states kept	6	Flagged by several methods but genuine

Table 9: Headline figures for Tab 5.

The 72 missing values are individual cells spread across the 34 columns with gaps identified in Section 3 (53 NFHS-5 indicators, each with Urban, Rural and Total columns).

What cleaning found and fixed (horizontal bar chart, univariate)

Chart Description: The chart shows the number of cells, columns, rows or states affected by each problem checked during cleaning, along with the steps taken to solve the issue.

Interpretation: Each bar represents one check, with its count and the action taken displayed at the end. The blue bars indicate problems that were fixed (empty columns dropped, text converted to numbers and missing values imputed), while the orange bars indicate checks that required no change: no duplicate rows were found and the outlier states were retained. Missing values are shown separately for the Urban, Rural and Total figures.

Rationale: A single bar chart provides an overview of the entire cleaning process. Distinguishing between problems that were fixed and checks that required no change demonstrates that all the checks were carried out, even when no correction was necessary.

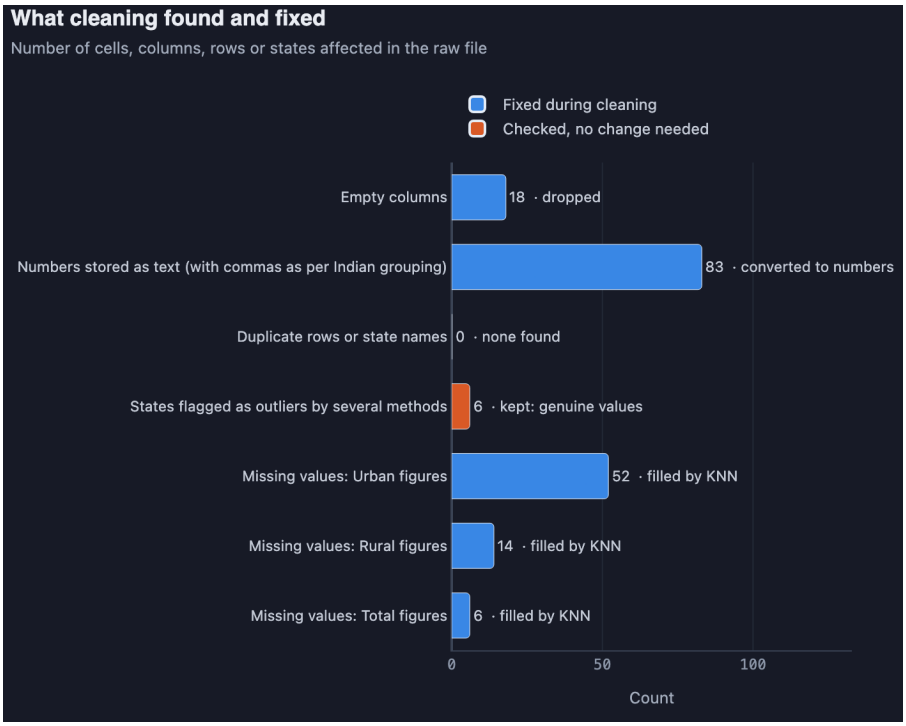


Figure 25: The raw file contained 18 empty columns, which were dropped, and 83 numbers stored as text with Indian-style commas, which were converted to numeric values. No duplicate rows were found. The 72 missing values were filled using KNN imputation. The 6 states flagged as outliers by more than one method were retained because their values are genuine.

Where the gaps were (heatmap, bivariate)

Chart Description: The chart shows the number of missing values in the raw data for each state, separated into Urban, Rural and Total figures.

Interpretation: Only the 14 states with at least one missing value are displayed, sorted from top to bottom in the descending order of the number of gaps. Each cell shows the number of missing values for that state and a set of figures. Lighter cells indicate more gaps, while blank cells indicate none.

Rationale: The missingno matrix in Section 3 shows missing values by column, whereas this heatmap shows them by state. It therefore attempts to answer whether missing data is concentrated in particular states. This is relevant when interpreting the results, since states with more imputed values may require greater caution in the other tabs.



Figure 26: Of the 72 missing values, 52 were in the Urban figures and only 6 were in the Total figures used for most of the analysis. Sikkim alone accounts for 35% of the gaps (25 values), most of them in the Urban figures. Small states with limited urban populations may have too few urban respondents for NFHS-5 to report a value.

Since only 6 of the 72 missing values were in the Total figures, imputation has limited influence on the results in Tabs 1–4, which use the Total figures unless stated otherwise.

Imputation fills the gaps without reshaping the data (split violin plot, univariate)

Chart Description: For the four Urban indicators with the most missing values, the chart compares the distribution across states before cleaning (using surveyed values only) with the distribution after KNN imputation.

Interpretation: Each panel represents one indicator, and its title gives the number of states for which values were imputed. Each violin is split into two halves: the left (orange) half shows the distribution before cleaning, while the right (blue) half shows the distribution post imputation. The width of a violin at a given height indicates how many states have values near that level. The horizontal line marks the mean. Hollow diamonds indicate the imputed values. Switching between Total, Urban and Rural figures can be carried out using the dropdown.

Rationale: Section 3 explains the reasoning behind KNN imputation and this chart provides a visual check of the results. A good imputation method should fill missing values with plausible estimates without substantially changing the overall distribution.

None of these four indicators is included in the Outcome Index or the Care Index. Hence, the imputation shown here does not directly affect either index.

Which states look unusual, and to how many methods (heatmap, multivariate)

Chart Description: The chart shows which states are flagged by each of the five outlier detection methods described in Section 3.

Interpretation: Each column represents a state (only the 24 states flagged by at least one method are shown) and each row represents a method. The three single-column methods (IQR, modified Z-score and Grubbs' test) show a number indicating how many of the 166 columns contain extreme values for

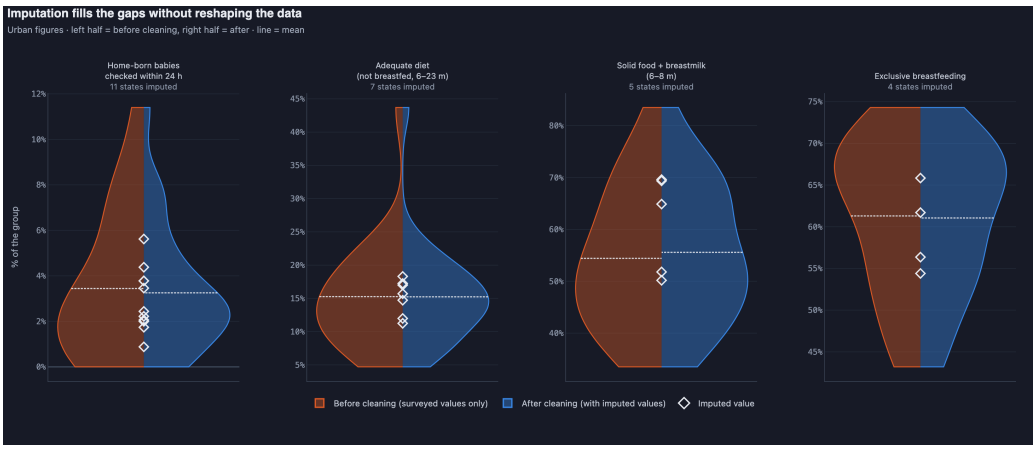


Figure 27: These four Urban indicators (home-born babies checked within 24 hours, adequate diet for non-breastfed children aged 6–23 months, solid food with breastmilk at 6–8 months, and exclusive breastfeeding) had 27 missing values between them. The distributions retain the same overall shape after cleaning, and the mean changes by at most 1.2 percentage points. All 27 imputed values lie within the range of the surveyed states, suggesting that imputation did not substantially distort the data.

that state. The two whole-profile methods (DBSCAN and Local Outlier Factor) show a tick mark if the state is flagged. Each row is shaded on its own scale, from dark purple to light pink, where light pink is the highest count for that method. Colours can therefore be compared within a row, while the printed numbers allow comparison between methods. States are ordered by the number of methods that flag them, with the states identified by the greatest number of methods appearing first on the left.

Rationale: A single test applied across 166 columns may flag some values by chance. Thus, it is necessary to check whether different methods identify those states as outliers. A heatmap of states against methods makes this agreement visible.

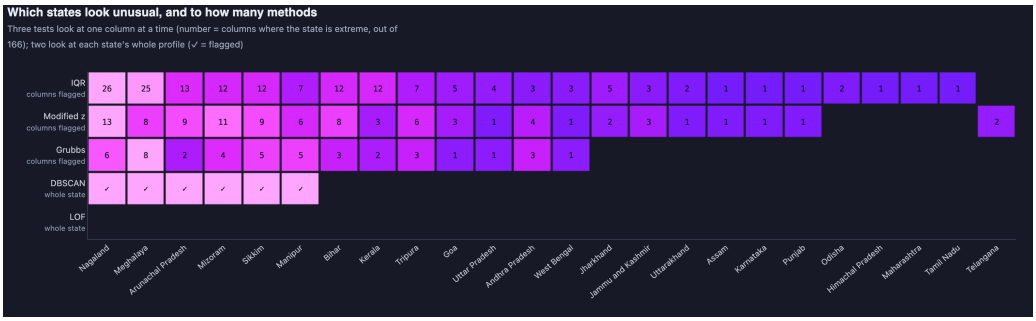


Figure 28: Of the 29 states, 24 are extreme on at least one column, which is not unexpected across 166 columns. Only the six small north-eastern states are flagged by both the whole-profile test (DBSCAN) and the single-column tests. Local Outlier Factor flags no state. Since their values are plausible and come from official government sources, no outliers were removed.

Retaining the six north-eastern states was a deliberate decision. Their influence on the results is also examined in Tab 4, where the main analysis is carried out, both with and without them. The association between spending and outcomes remains ($\rho = +0.68$, $p < 0.001$, $n = 23$).

Summary of Tab 5

The raw data required only a limited number of corrections: empty columns were dropped, text values were converted to numbers, and 72 missing cells were imputed. Most missing values occurred in the Urban figures of a few small states, while only 6 were in the Total figures used for most of the analysis. The imputed values remain within the range of the surveyed data. The six north-eastern states were flagged by several outlier tests, but their values were considered genuine and retained. Their influence on the findings was examined separately in Tab 4.

4.11 Tab 6: Findings and Methods

Key findings

Table 10 lists the seven findings presented on the dashboard, along with the figures in this report that support each one.

#	Finding	Evidence	Where
1	Higher health spending is associated with better health outcomes	Strong association with the Outcome Index ($\rho = +0.77$, $p < 0.001$); all 10 individual outcomes are significantly better in higher-spending states, including infant mortality ($\rho = -0.57$).	Tab 3; Figs. 16, 17
2	Higher spending does not necessarily mean greater use of healthcare	No clear association with the Care Index ($\rho = -0.08$, $p = 0.68$); only 1 of 9 care indicators is significant. Without the six north-eastern states, $\rho = +0.44$.	Tabs 3, 4; Figs. 20, 23
3	Other factors partly explain the relationship	The association remains significant after adjusting for each of the 5 factors; the lowest adjusted value is $\rho = +0.52$, after allowing for women's literacy.	Tab 4; Fig. 22
4	The association persists across alternative samples	$\rho = +0.68$ without the north-eastern states and $\rho = +0.76$ for the 18 states surveyed before COVID-19.	Tab 4; Fig. 23
5	Higher public spending does not necessarily reduce household costs	The out-of-pocket cost of a delivery in a public facility increases with spending ($\rho = +0.70$, $p < 0.001$).	Tabs 3, 4; Fig. 24
6	States achieve different outcomes at similar spending levels	Kerala scores 33 points above its predicted Outcome Index and Gujarat 20 points below; 22 of 29 states spent less than budgeted in 2019-20.	Tabs 1, 3; Figs. 19, 10
7	Several states fall below the natural sex ratio at birth	12 of 29 states have fewer than 935 girls per 1,000 boys at birth; no clear association with spending ($\rho = +0.27$, $p = 0.16$).	Tab 2; Fig. 15

Table 10: Key findings shown in Tab 6 and the figures that support them.

How the analysis was done

The methods panel in Tab 6 summarises the steps described in detail in Sections 2 to 4. These include the data sources, data-cleaning procedures, calculation of per capita spending, construction of the two indices and statistical methods, including Spearman's ρ, partial correlations and the two robustness checks. This summary allows readers who have not seen the report to understand how the dashboard's findings were obtained.

Limitations

The limitations panel in Tab 6 identifies eight limitations of the analysis. Each is listed below, along with the steps taken to reduce its effect where possible.

1. **Correlation does not imply causation.** The analysis compares 29 states at one point in time. Factors such as income, education, governance and infrastructure may influence both spending and outcomes. *Mitigation:* five possible confounders were examined using partial correlations (Tab 4), but unmeasured factors, particularly state income, could not be tested.
2. **Health expenditure coverage is limited.** The spending measure includes only revenue expenditure by state governments on Medical and Public Health. It excludes capital expenditure, central government and private spending, as well as expenditure recorded under other budget heads such as Family Welfare, which also funds some maternal and child health services.

3. **Survey timing differs across states.** NFHS-5 was conducted in two phases, with Phase 2 overlapping with the COVID-19 pandemic. *Mitigation:* the main result was recalculated for the 18 Phase 1 states and remains almost unchanged ($\rho = +0.76$ compared with $+0.77$ for all states).
4. **Maternal mortality is not directly measured.** NFHS-5 does not report maternal mortality, and Sample Registration System estimates are not available separately for all 29 states. Maternal health is therefore assessed indirectly using indicators such as anaemia during pregnancy and women's nutrition.
5. **Imputed values introduce uncertainty.** A total of 72 missing values were imputed, 52 of them in the Urban figures. *Mitigation:* only 6 missing values were in the Total figures used for most of the analysis. Imputed values are marked with hollow markers in the charts, and Tab 5 shows that imputation did not substantially change the distributions.
6. **State estimates vary in precision.** NFHS-5 estimates for small states, particularly their Urban figures, are based on smaller samples. *Mitigation:* the six small north-eastern states were examined separately in Tab 4.
7. **Sex ratio at birth is imprecise for small states.** For Goa, NFHS-5 reports 838 girls per 1,000 boys, compared with 947 in the Civil Registration System for 2020. Himachal Pradesh shows a similar difference (875 versus 936). Therefore, state rankings on this indicator should be interpreted with caution.
8. **Geographical coverage is restricted.** The analysis covers 29 states (counting Jammu and Kashmir) and excludes the remaining Union Territories, including Delhi and Ladakh. The findings therefore apply to the states included in the analysis.

Two additional limitations arise from the design of the analysis itself:

9. **Only one year of spending is used.** Health outcomes such as stunting and infant mortality reflect investments made over several years, whereas this analysis uses spending data for a single year (2019-20). Spending in that year is therefore treated as an indicator of each state's general spending level, based on the assumption that the relative spending levels of states change slowly. This assumption was not tested.
10. **The analysis uses state-level data.** All comparisons are based on state averages. The observed relationship between states does not necessarily hold for individual families within a state. The indices are also relative: a score of 100 represents the best-performing state among the 29, not an ideal level of performance. The results also depend on the use of equal weights when constructing the indices (Section 4.2).

5 Conclusion

This project examined whether Indian states that spend more per capita on health have better maternal and child health outcomes. Using RBI state health expenditure data for 2019-20 and NFHS-5 indicators for 29 states, the findings show a clear positive association. States that spend more per capita tend to have better health outcomes. The correlation between per capita spending and the Outcome Index is strong ($\rho = +0.77$, $p < 0.001$), and all 10 individual health outcomes are significantly better in higher-spending states. This association remains significant after adjusting for each of the five possible confounders (with partial correlations ranging from $+0.52$ to $+0.80$), excluding the six north-eastern states ($\rho = +0.68$), and considering only the states surveyed before the COVID-19 pandemic ($\rho = +0.76$).

Three additional findings provide further context to this result:

- **Spending is not clearly linked to the care received.** Across all 29 states, the Care Index shows no clear association with spending ($\rho = -0.08$). This pattern is largely influenced by the north-eastern states, which have high per capita spending but relatively low use of formal

healthcare. Without these states, care increases with spending ($\rho = +0.44$). The relationship between spending and better outcomes is therefore not simply a matter of "more money, more antenatal visits and facility births".

- **Public spending does not appear to reduce costs for families.** Families pay more, rather than less, for a delivery in a public facility in higher-spending states ($\rho = +0.70$). One possible explanation is that healthcare costs are higher in remote and hilly states.
- **Spending is not the whole story.** States with similar spending levels achieve very different outcomes. Kerala scores 33 points above the level predicted by its spending, while Gujarat scores 20 points below. Women's literacy and sanitation also account for part of the association between spending and outcomes.

Since the data are cross-sectional and observational, these findings establish an association, not a causal relationship. They are consistent with health spending contributing to better maternal and child health, but differences between states that were not measured, such as income, may also help explain the results. Future work could strengthen the analysis by using several years of spending data to examine changes within states over time, including Family Welfare and capital expenditure, adding state income as a control variable, and using district-level NFHS-5 data to increase the number of observations.

The contributions of each team member and the use of AI tools are documented in `contribution.pdf` and `ai_use.pdf` files.